

Microbiology

World of Revision

Marrow

MARROW

Instructions

- Notes are to be used in conjunction with Marrow videos.

Please note:

- The information in this book has been printed based on the transcript of the Marrow videos. This book has to be used in conjunction with the Marrow videos and not as a standalone material.
- The information contained in this book is for educational purposes only. The content provided is not intended to substitute for professional medical advice, diagnosis or treatment.
- This book cannot be sold separately. It has been made available to only select eligible users who have an active subscription to Marrow videos.
- The text, images, slides, and other materials used in this book have been contributed by the faculty, who are subject matter experts. We have merely reproduced them as video transcripts in this book.
- The notes have been consciously designed in a way that is concise and revisable. To ensure this, we have intentionally added only the most relevant modules and images that are needed for you.
- The notes contain blank spaces primarily for labelling diagrams, completing cycles and more to promote active engagement and reinforce learning.
- Red icons, wherever present, serve as cues to faculty-emphasised sections, intended to guide focused learning.
- Reasonable care has been taken to ensure the accuracy of the information provided in this book. Neither the faculty nor Marrow takes any responsibility for any liability or damages resulting from applying the information provided in this book.

All Rights Reserved

No part of this publication shall be reproduced, copied, transmitted, adapted, modified or stored in any form or by any means, electronic, photocopying, recording or otherwise.

Contents

Microbiology

General Microbiology : Part 1	1
Pioneers in microbiology • Morphology & motility • Culture media (CM) • Biochemical tests • Molecular methods • Immunological methods • Antimicrobial susceptibility test	
General Microbiology : Part 2	8
General bacteriology • General mycology	
General Microbiology : Part 3	11
General virology • General parasitology	
Hospital Infection Control	17
Hand hygiene & PPE • Biomedical waste management (BMW) • Sterilization • Disinfection	
Immunology	20
Immunity • Antigen • Antibody • Hypersensitivity reactions • Transplant immunology • Immunodeficiency disorders (IDD) • Complement deficiency disorders (CDD)	
Central Nervous System	25
Bacterial Meningitis • Viral encephalitis • Parasitic encephalitis • Viral myelitis • Bacterial Neurotoxin related diseases	
Cardiovascular System : Infections of Heart	30
Infective endocarditis • Acute rheumatic fever (ARF)	
Cardiovascular System : Bloodstream and Lymphatic Infections	33
HIV • Viral hemorrhagic fever (VHF) • Blood and tissue flagellates • Blood and tissue sporozoa • Helminths	
Respiratory System	42
Diphtheria, pertussis and TB • Typical/lobar pneumonia • Atypical/interstitial pneumonia : Bacterial, viral, parasitic • Fungal atypical pneumonia	
Gastrointestinal & Hepatobiliary System	49
Peptic ulcer disease • Bacterial gastroenteritis • Viral gastroenteritis • Parasitic gastroenteritis • Helminthic gastroenteritis • Viral hepatitis • Parasitic hepatitis • Hydatid cystic disease	
Genitourinary System	55
Diseases of genital system • Diseases of urinary system	
Skin, Subcutaneous & Musculoskeletal System	57
Bacterial infections • DNA viruses • RNA viruses • Superficial fungal infections • Subcutaneous fungal infections • Parasitic infections	
Miscellaneous Infections	64
Bacterial toxin related diseases • Bacterial zoonotic diseases • Bacterial vector transmitted diseases • Congenital infections • Prion disease • Bioterrorism agents & biosafety levels	

GENERAL MICROBIOLOGY : PART 1

----- Active space -----

Pioneers in Microbiology

00:07:32

	Contribution	Significance
Antonie van Leeuwenhoek	Simple microscope	Father of microscopy
Louis Pasteur	- Liquid culture media - vaccines (CAR) : Cholera, Anthrax, Rabies	Father of microbiology
Robert Koch	- Solid culture media - Discovered (CAT) : Cholera, Anthrax, TB	Father of bacteriology
Kary B mullis	molecular methods (PCR)	-
Eva Engvall (Along with Peter Perlman)	Immunological methods (ELISA)	-

Morphology & Motility

00:16:45

Microscopy

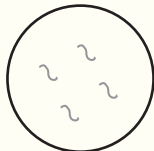
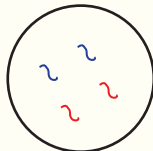
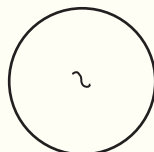
morphology detection :

- Primarily : Capsule, cell wall, fimbriae.
- methods : Wet mount, staining.

motility detection :

- Primarily : Corkscrew motility.
- method : Hanging drop.

----- Active space ----- Types of microscopes :

	Bright field/light microscope (m/c)		Dark field microscope	Phase contrast microscope
	Wet mount/ Hanging drop	Staining		
morphology	Visualised	Visualised better	Visualised : No staining	
motility	Appreciated	Not appreciated	Corkscrew motility : Spirochetes	
Appearance			Thin organisms	
	Bright organisms with bright background	Colourful organisms with bright background		
Images				

Capsule Detection :

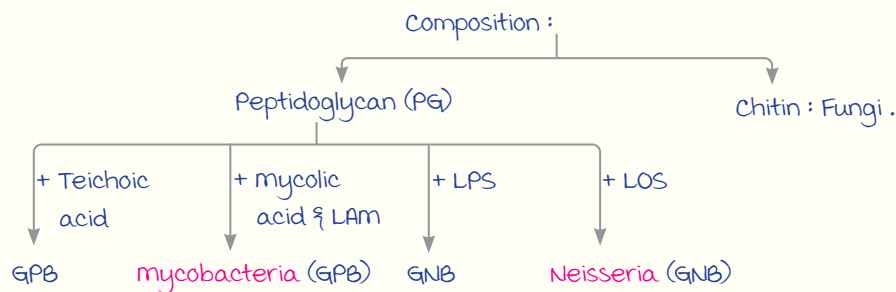
	microscopic method		Immunological method	
	Negative staining (Screening test)	McFadyen's staining	Latex agglutination test (Best/rapid test)	Quellung's reaction
MOA	Capsule staining		Detects capsular Ag	
Indicator	India ink/nigrosin	Polychrome methylene blue	Clumps	Capsular swelling
Capsule type detected	All types	Only polypeptide (B. anthracis)	All types	Only polysaccharide (Pneumococcus, meningococcus, Hib)

Cell Wall Detection :

All bacteria except mycoplasma (mollicutes) have cell wall.

Identification :

----- Active space -----



GPB : Gram positive bacteria
GNB : Gram negative bacteria
LAM : Lipoarabinomannan
LPS : Lipopolysaccharide
LOS : Lipooligosaccharide

Fimbriae/Pili Detection :

- Organ of attachment.
- UTI bacteria (m/c : E.coli).

Identification : Heme agglutination.

Culture Media (CM)

00:34:25

Types :

Simple CM : Growth of all organisms.

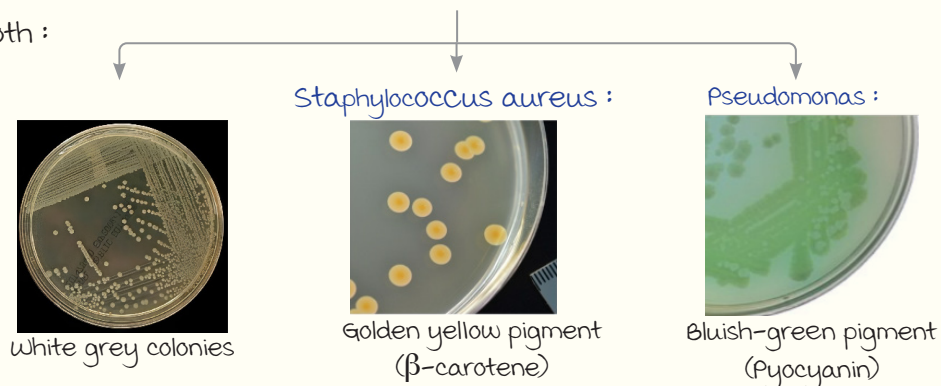
Special CM :

- Selective CM + Special substances.
- Types :

	MOA
Enriched media	⊕ Growth of pathogens
Enrichment media	⊖ Commensals in stool samples
Differential media	Differentiates organisms
Transport media	Keep organism alive during transport
Selective media	⊕ Growth of specific organisms
Anaerobic media	⊕ Growth of anaerobic organisms

Simple/Basal CM :

Nutrient agar/broth :

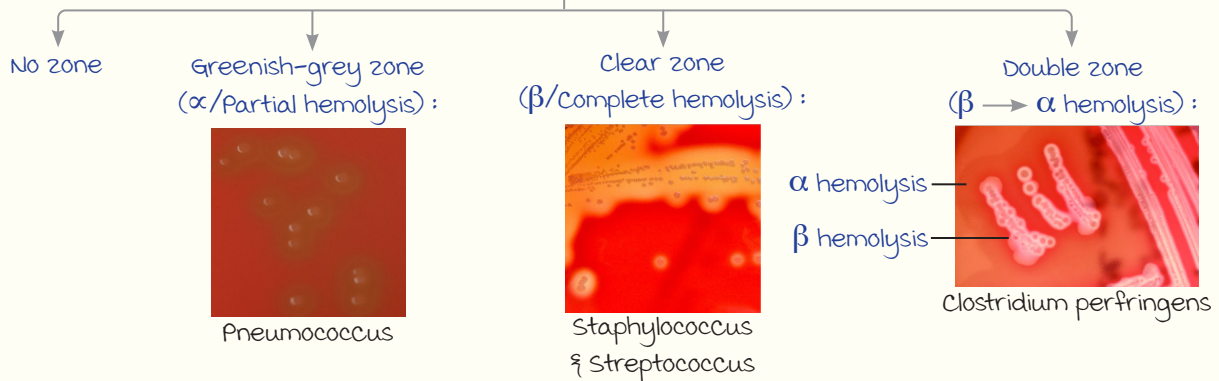


----- Active space -----

Enriched media :

Blood agar :

Observation :

Enrichment media :

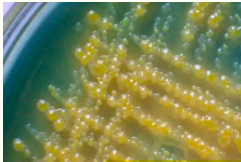
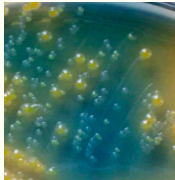
	Selenite F broth	Tetrathionate broth	Alkaline peptone water
Colour	Light brown	Bluish green.	Straw coloured
Pathogens grown	All stool pathogens (Except : <i>Vibrio</i>)		<i>Vibrio</i>

Note :

Sputum sample : NALC - NaOH method (NALC : N-Acetyl-L-cysteine).



Differential media :

		macConkey agar	CLED agar
Composition		Lactose sugar	
		Bile salts ($\ominus$ GP)	Cysteine-lactose-electrolyte deficient
Organisms		GN	GN/GN, GP, Candida
Observation	Lactose fermenters : • <i>E. coli</i> • <i>Klebsiella</i>	 Pink colonies	 Yellow colonies
	Non-lactose fermenters	many organisms  Pale colonies	 Blue colonies

Transport media (Stool) :

----- Active space -----

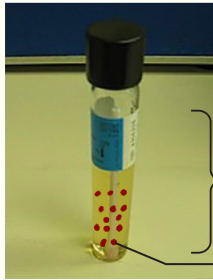
Cary Blair media :

All pathogens

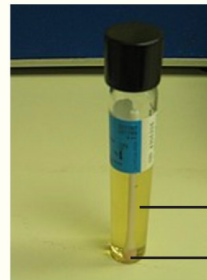
Buffered Glycerol Saline :

All pathogens except *Vibrio* &
*Campylobacter*Anaerobic media :

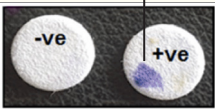
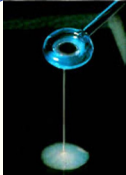
Robertson cooked meat (RCM) :

Only anaerobic
growth
meat
particles

Thioglycollate broth :

Aerobic growth
Anaerobic growthBiochemical Tests

00:52:24

	Observations	Organisms
Catalase test	Bubbles  +ve -ve	Catalase -ve : • Streptococcus • Pneumococcus • Enterococcus
Oxidase test	Blue colour  -ve +ve	Oxidase +ve : • Vibrio • Pseudomonas • Campylobacter • Helicobacter
urease test	Pink colour : +ve	Urease +ve : • Helicobacter • Proteus • Nocardia
Coagulase test	Coagulum  -ve +ve	Coagulase +ve : Staphylococcus aureus
String test	String formation : +ve 	String test +ve : Vibrio cholerae

----- Active space -----

Molecular Methods

00:56:40

Polymerase Chain Reaction (PCR) : 🔗

Detects gene.

Steps :

NA extraction : Lysozyme



NA amplification : Thermocycler [Denaturation (95°C) → Annealing (55°C) → Extension (72°C)] → 40 cycles

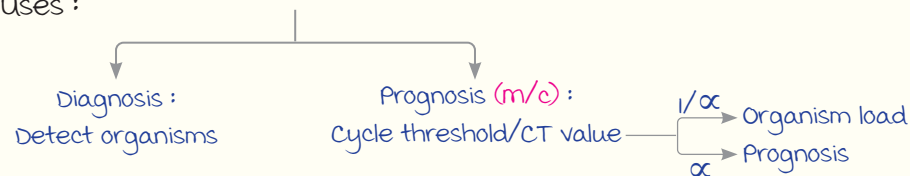

 NA detection : Gel electrophoresis
fluorescent method.

NA : Nucleic acid

modifications :

	Characteristics
Conventional PCR	Detects only DNA
Real time RT PCR	• Detects DNA & RNA • Uses reverse transcriptase (RT)
Multiplex realtime RT PCR	• Detects multiple organisms • For syndromic approach
Automated realtime RT PCR	• Cartridge based : CBNAAT • Chip based : TruenAT

Uses :

**Immunological Methods** 🔗

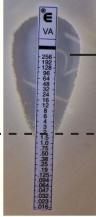
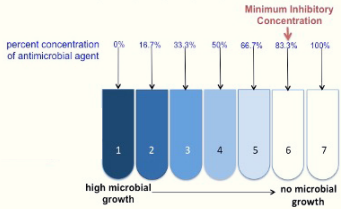
01:05:30

	Tests	Organisms
Ag detection	Nagler's reaction	Clostridium perfringens
	Elek's gel precipitation test	Corynebacterium diphtheriae
Ab detection	Widal agglutination test (WAT)	Salmonella
	Standard agglutination test (SAT)	Brucella
	microscopic agglutination test (MAT)	Leptospira
	Weil Felix test	OX K Ag +ve Orientia
		OX 2 ± OX 19 Ag +ve Rickettsia

Antimicrobial Susceptibility Test

01:09:00

----- Active space -----

	Tests	medium	Technique	Image
Phenotypic methods (Culture media)	Kirby bauer disc diffusion (m/c)	mueller-Hinton agar	- ZOI $\oplus$: Sensitive $\ominus$: Resistant - MIC not obtained	
	E-strip	mueller-Hinton agar	- ZOI $\oplus$: Sensitive $\ominus$: Resistant - MIC obtained	
	Broth dilution (Gold standard)	mueller-Hinton broth	- Test tube turbidity $\ominus$: Sensitive $\oplus$: Resistant - MIC obtained	
Genotypic methods (molecular)	PCR 	-	- Resistant gene $\ominus$: Sensitive $\oplus$: Resistant - m.tb $\xrightarrow{\text{CBNAAT}}$ $\text{rpo-B gene } \oplus$ Rifampicin resistance	-

ZOI : Zone of inhibition

MIC : minimum inhibitory concentration

----- Active space -----

GENERAL MICROBIOLOGY : PART 2

General Bacteriology

00:00:26

Classification :

G⁺ve cocci :

- Staphylococcus.
- Streptococcus.
- **Pneumococcus.**
- Enterococcus.

G⁻ve cocci :

- **meningococcus.**
- Gonococcus.

G⁺ve bacilli :

- Bacillus.
- **Clostridium.**
- Corynebacterium.
- **mycobacterium.**
- **Actinomyces.**
- Nocardia.
- Listeria.

G⁻ve bacilli

Enterobacteriaceae :

- Escherichia.
- **Klebsiella.**
- Salmonella.
- Shigella.
- Yersinia.
- Proteus.
- Citrobacter.

Oxidase +ve :

- Vibrio.
- **Pseudomonas.**
- Campylobacter.
- Helicobacter.
- Hemophilus (**Hib**).
- Brucella.
- Bordetella.

Spirochaetes :

- Treponema.
- Borrelia.
- Leptospira.

Non-cultivable :

- **Rickettsia.**
- **Chlamydia.**
- T. pallidum
- m. leprae

Atypical pneumonia :

- Chlamydia.
- mycoplasma
- Legionella.

miscellaneous :

- HACEK group.
- Acinetobacter.
- **Gardnerella vaginalis.**

Key :

- Obligate intracellular
- Facultative intracellular : **Neisseria**
- Capsulated bacteria
- Obligate anaerobes
- Facultative anaerobes : without O₂ preferred

Safety pin appearance :

Wayson's/bipolar staining :

Genital ulcer :

- **Klebsiella granulomatis.**
- **Haemophilus ducreyi.**

Pneumonia :

- **Burkholderia pseudomallei.**
- **Yersinia pestis.**

Gastroenteritis :

- **Vibrio parahaemolyticus.**

Lab Diagnosis :

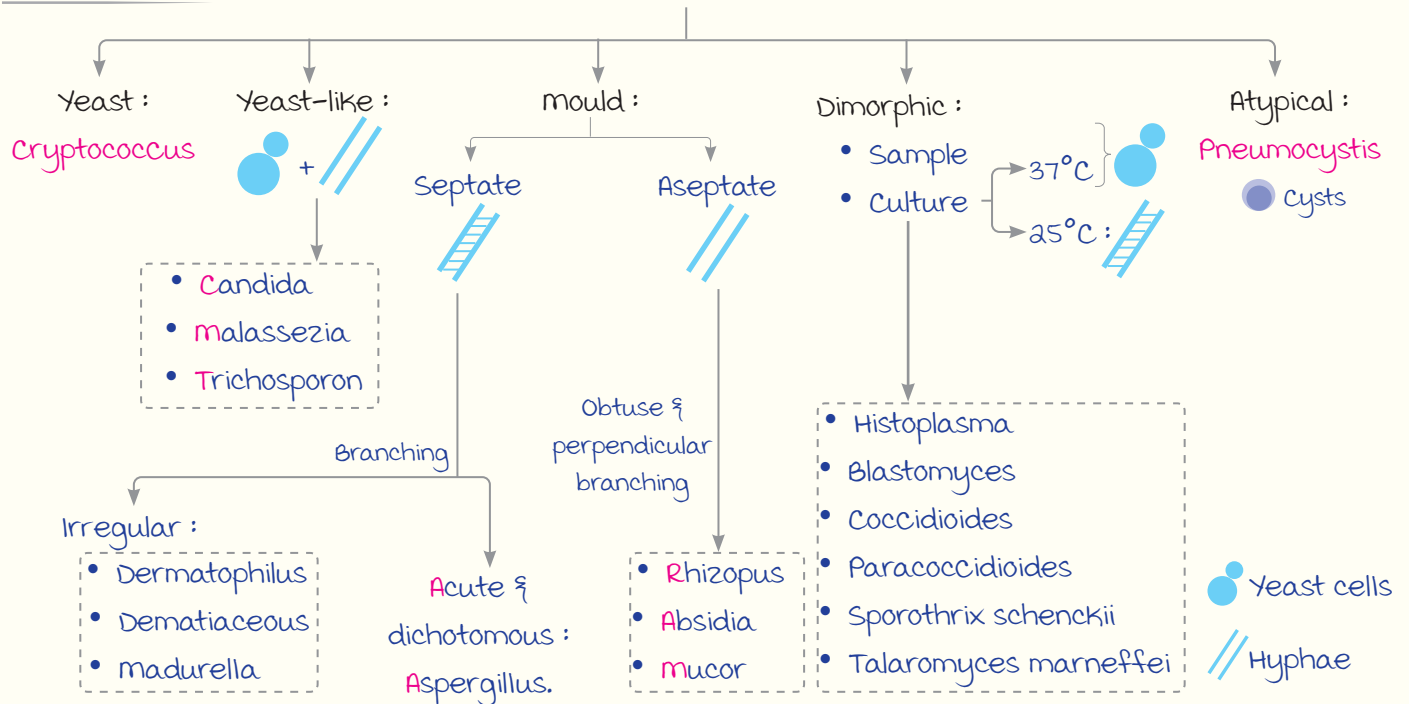
----- Active space -----

Cultivable bacteria : **microscopy** → **Culture & biochemical test.**

Non-cultivable bacteria
Slow grower bacteria (m. tb) } **microscopy** → **molecular/immunological tests.**

General Mycology

00:29:21

Classification :**Lab Diagnosis :**

		Yeast	Yeast-like	mould	Dimorphic	Atypical
Sample microscopy		⊕				
SDA (Spores)	Normal	⊖		⊕	⊕ (25°C)	Non-cultivable
	Special condition	⊕				
Culture test/ microscopy		Test		microscopy		
Confirmatory tests		molecular ± Immunological test				molecular test : pcr

SDA : Sabouraud dextrose agar

----- Active space -----

Spores 🦠:

		Produced by
Sexual	Ascospores	Septate mould
	Basidiospores	Yeast & yeast-like (except Candida)
	Zygospores	Aseptate mould
Asexual	Arthrospores	- Dermatophyte - Coccidioides - Trichosporon
	Blastospores	Yeast & yeast-like
	Chlamydospores (Special condition)	Candida
	Conidiospores	Septate mould
	Sporangiospores	Aseptate mould

Invasive markers 🦠:

β -D-Glucan	Galactomannan
- Candida - Aspergillus - Pneumocystis	Aspergillus

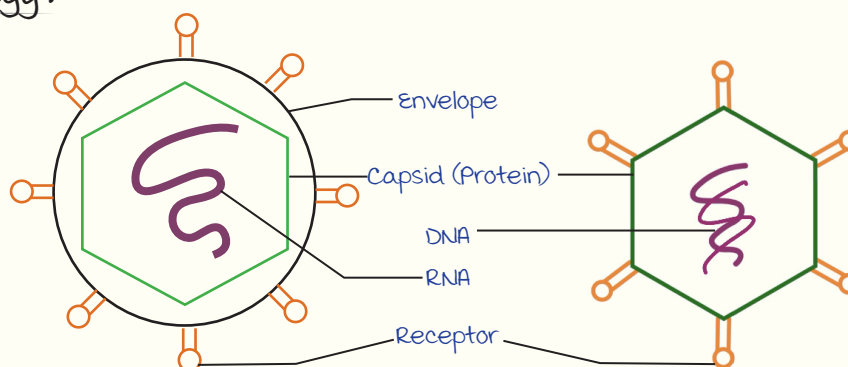
GENERAL MICROBIOLOGY : PART 3

----- Active space -----

General Virology

00:00:15

morphology :



Classification of virus :

DNA viruses 🦠 :

Enveloped (3) (HHP)	Non-enveloped (4) (PPAP)	
Hepadna (HBV)	Papilloma (HPV)	Adenovirus
Herpes (HHV)	Polyoma (JC, BK)	Parvovirus
Pox virus		

RNA viruses 🦠 :

		Virus
Enveloped (13)	Retro	HIV
	Flavi	- HCV - JE - YF - KFD - Dengue - West Nile - Zika
	Toga	Chikungunya
	matona	Rubella
	Arena	Lassa fever v.
	Bunya	- Hanta v. - CCFV
	Orthomyxo	Influenza v.
	Paramyxo	- Parainfluenza - measles - Nipah - mumps
	Pneumo	- RSV - hmpv
	Rhabdo	- Rabies v. - Chandipura v.
	Filo	- Ebola v. - marburg v.
	Delta	HDV
	Corona	Corona v.
Non enveloped (5) (PC - HAP)	Picorna	- HAV - Polio v. - Coxsackie v. - Enterov.
	Calci	- Noro v. - Sapo v.
	Hepe	HEV
	Astro	Astro v.
	Reo	Rota v.

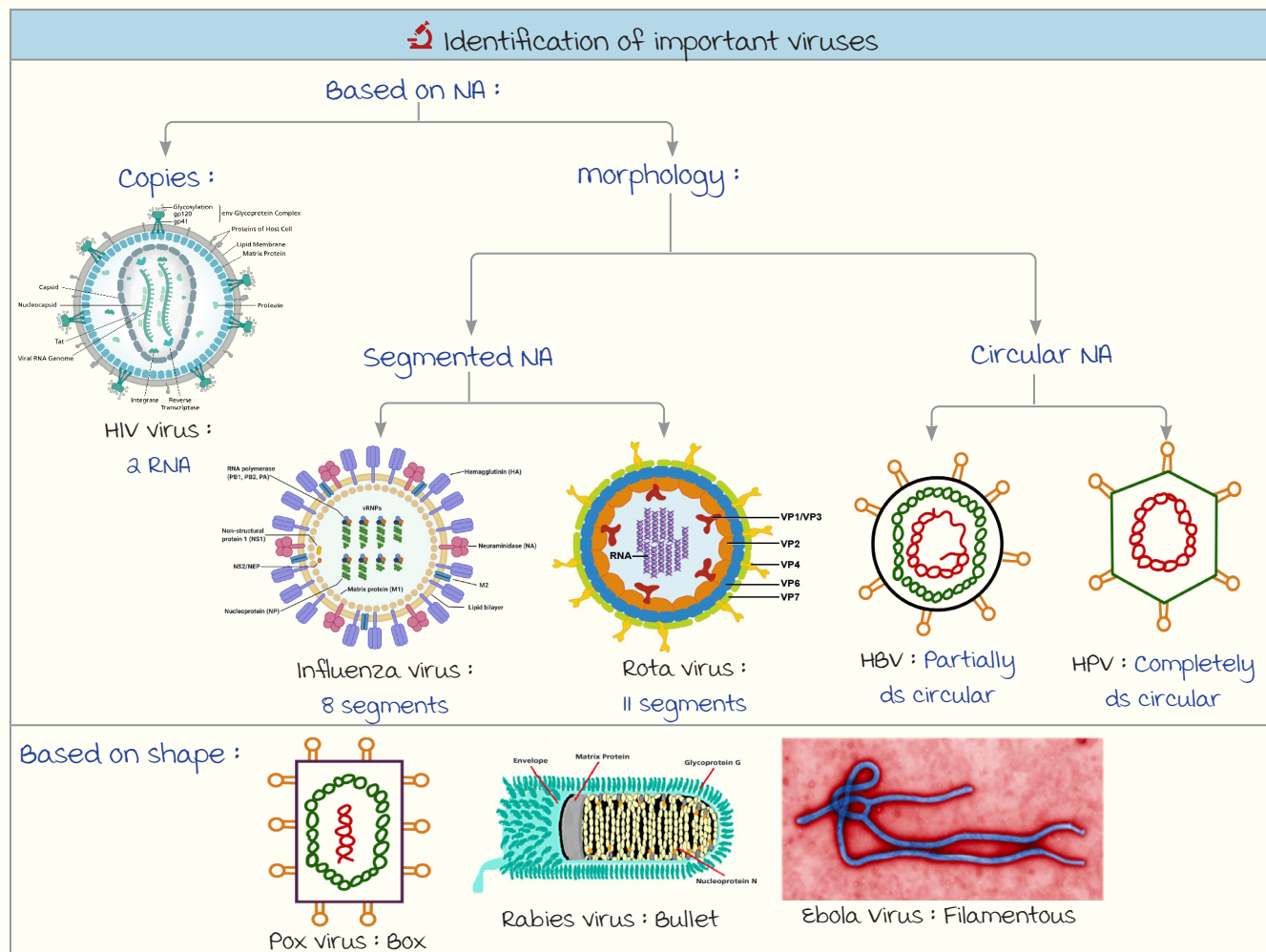
Paradox NA viruses

- ss DNA : Parvo v.
- ds RNA : Rota v.

Imp viruses : Size

- Biggest : Pox
- Smallest : Parvo

----- Active space -----



Laboratory Diagnosis :

microscopy

- Electron microscope : Viruses visualised.
- Bright field microscope : Inclusion bodies visualised.

Culture :

- Research :
 - Animal inoculation.
 - Embryonated egg culture.
- Lab diagnosis :
Tissue/cell culture (m/c).

Live cells.

Intranuclear	Intracytoplasmic
• HSV : Cowdry type A/ Lipschütz • CMV : Owl eye • YFV : Torres/Councilman	Rabies V : Negri

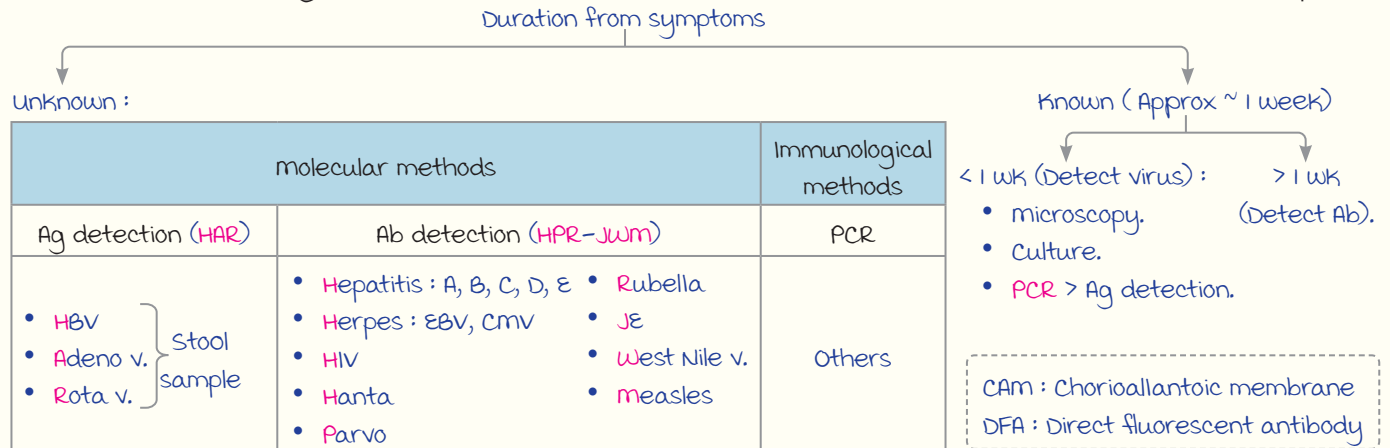
Culture identification :

- DFA : m/c
- Cytopathic effect :

	Cytopathic effect
measles v.	Syncytium formation of cells
Adenovirus	• Granular clumping • Bunch of grapes
HSV	Ballooning of cells
Enterovirus	Crenation of cells

molecular & immunological methods 📌 :

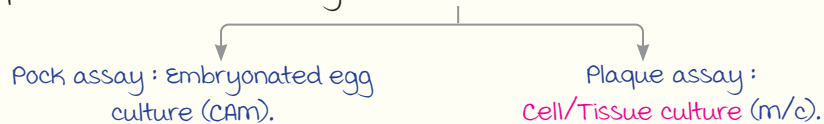
----- Active space -----



Note :

Dengue → < 5 d : Detect virus.
→ ≥ 5 d : Detect Ab.

virus quantification/viral assays :

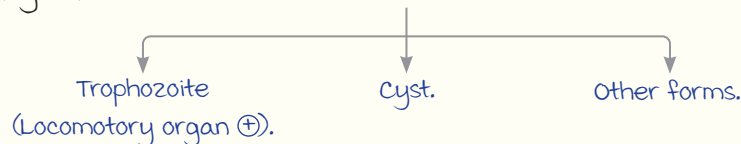


General Parasitology

00:48:37

General Protozoology :

morphological forms :



Locomotory classification 📌 :

		Ciliate	Amoeba	Flagellates	Sporozoa/coccidia
Locomotory organ		Cilia	Pseudopodia	Flagella	⊖
Sites	Intestine	Balantidium coli	Entamoeba histolytica	Giardia lamblia	• Cryptosporidium • Cyclospora • Cystoisospora
	Blood & tissue	-	-	• Leishmania • Trypanosoma	• Plasmodium • Babesia • Toxoplasma
	Others	-	Free living (Soil & water) : • Balamuthia • Naegleria • Acanthamoeba • Sappinia	Vaginal : Trichomonas vaginalis	

----- Active space -----

Opportunistic protozoa
• Cryptosporidium : Autoinfection • Cyclospora • Cystoisospora • Giardia • Toxoplasma
Acid fast

Laboratory diagnosis 📌:

- **microscopy** : Preferred

Protozoa	Sample	Observation
Intestinal	Stool	Trophozoite & cyst
Free living	CSF	Trophozoite ± cyst
Vaginal	Vaginal discharge	Only trophozoite
Blood & tissue	• PBS • Tissue biopsy	Other forms

- Immunological tests 📌:

	Test	Principle
Leishmania	Rapid test	Detects Ab against rk 39 Ag
Plasmodium Vivax		PLDH Ag
Falciparum		PLDH Ag + HRP-2 Ag
Toxoplasma	Sabin Feldman dye test	Detects IgG Ab

General Helminthology :

morphological forms :

	Worms	Egg	Larvae
Sample	Ruptured blister : Dracunculus medinensis	Stool	• Stool • PBS • muscle tissue

morphological classification of worms :

----- Active space -----

		Cestodes	Trematodes	Nematodes
Shape		Flat		Round
AKA		Tapeworms	Flukes	Roundworms
Site	Intestine	- Taenia solium : Pork - Taenia saginata : Beef - Hymenolepis nana : Dwarf - Diphyllobothrium latum : Fish	Fasciola buski	Strongyloides stercoralis : Parthenogenic - Ancylostoma duodenale : Hook - Ascaris lumbricoides : Round - Enterobius vermicularis : Pin - Trichuris trichiura : Whip
	Tissue	Echinococcus granulosus : Dog	- Schistosoma : Blood - Fasciola hepatica - Clonorchis sinensis - Paragonimus westermani : Lung } Liver	Filarial worm Trichinella spiralis : Spiral Dracunculus medinensis : Guinea

: Ovoviviparous

: Viviparous

Others : Oviparous

Lab diagnosis :

microscopy (Preferred) 🦋 :

	Sample	Visualize
Oviparous helminths	Stool	Egg
Ovoviviparous helminths		Larvae
viviparous helminths	- PBS - muscle biopsy	Larvae
	Ruptured blister	Adult worm

🦋 Eggs	
Non-bile stained	Float in salt solution
- Necator - Enterobius H. nana - Ancylostoma	- Trichuris - Enterobius H. nana - Ancylostoma - Necator - Ascaris

----- Active space -----

Pathogenesis :

Infection by :

🔪 Eggs		🔪 Skin penetration	
Cestodes	Nematodes	Trematode	Nematodes
• Dog tapeworm • Dwarf tapeworm	• Ascaris • Trichuris • Enterobius	Schistosoma	• Strongyloides • Ancylostoma

🔪 Autoinfective	
Cestodes	Nematodes
• T. solium • H. nana	• Enterobius • Strongyloides

(Opportunistic)

Larva migrans (Lm) 🔪 :

	Human Nematodes	Animal Nematodes (m/c)
visceral Lm	Ascaris lumbricoides	Toxocara canis
Cutaneous Lm	• Strongyloides stercoralis (Larva currens) • Ancylostoma duodenale	• Ancylostoma braziliense • Ancylostoma caninum

Loeffler's syndrome/Loeffler's pulmonary eosinophilia
(Human > Animal nematode)

Anemia 🔪 :

	Helminths
megaloblastic	D. latum
Iron deficiency	• Ancylostoma/Necator • Trichuris

Cancer 🔪 :

	Helminths
Bladder Ca	Schistosoma haematobium
Colon Ca	Schistosoma japonicum
Cholangio Ca	• Clonorchis • Opisthorchis

HOSPITAL INFECTION CONTROL

----- Active space -----

Hand Hygiene & PPE

00:01:01

Hand Hygiene :

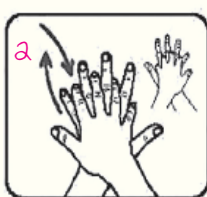
methods :



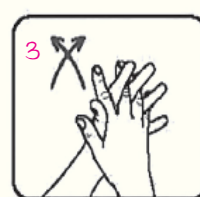
WHO steps of hand hygiene :



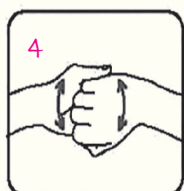
Rub hands **palm** to palm.



Rub hand's **back** with other palm, fingers interlaced.



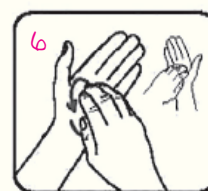
Rub palm to palm, fingers interlaced.



Rub backs of fingers to opposing **palms** (knuckles to palm).



Rub each **thumb** with rotational motion.



Rub tips of fingers onto palm in a circular motion.

5 moments for hand hygiene :

- Before & after touching patient.
- Before & after procedures & fluid exposure.
- After touching patient surroundings.

Personal Protective Equipment (PPE) :

Two methods :

	1. Donning	2. Doffing
Action	Putting PPE on	Removing PPE
Sequence	Hand hygiene → Gown → mask → Goggles/ Face shield → Gloves .	Gloves → Gown → Goggles/Face shield → mask → Hand hygiene

----- Active space -----

Biomedical Waste Management (BMW)

00:10:16

Segregation & Disposal methods :

	Yellow bag	Red bag	White bag	Blue bag
Type of waste	- Soiled cotton/Linen - Tissues, culture media - Chemicals, medicines - Blood bags (Plastic)	- Plastic : urine bag - Rubber : Soiled gloves	- metallic sharps - Syringes with inseparable needle	- metals - Glass
Disposal method	Incineration	Autoclave ↓ F/b Recycling	Na hypochlorite ↓ F/b Shredding/Pack in puncture proof containers then bury (Encapsulation).	Na hypochlorite ↓ F/b Recycling.

mercury :

Aspirate in syringe → Store in container with water → Handover to BMW collectors.

Sterilization

00:21:23

Sterilization vs. Disinfection :

Complete killing of all forms of microorganisms — { Including spore form : Sterilization.
 Excluding spore form : Disinfection.

Types of Sterilization :

	moist heat	Dry heat	H ₂ O ₂ /Plasma	Ethylene oxide (ETO)	Ionising radiation
Example	Autoclave	Hot air oven	-	-	γ rays
m/c method	121°C for 15 min under 15 psi	160°C for 2 hours	-	-	-
Control	Bacillus stearothermophilus	Bacillus subtilis/ Bacillus atrophaeus	Bacillus stearothermophilus	Bacillus subtilis/ Bacillus atrophaeus/ Clostridium tetani	Bacillus pumilus

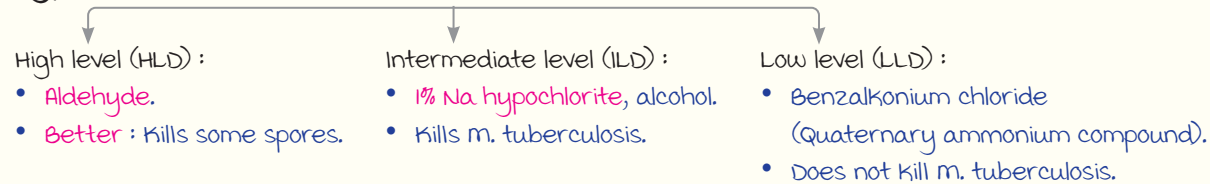
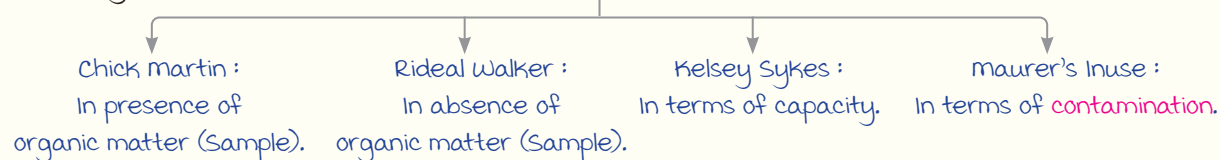
materials Sterilized 📌:

----- Active space -----

materials	method
- Surgical instruments except sharps (E.g. : Linen, sutures without needle) - Rigid endoscopes (Laparoscope, arthroscope)	Autoclave > H_2O_2 > ETO
Plastic & rubber materials (E.g. : Syringe, gloves)	Autoclave > Ionising radiation
Glass & sharps (E.g. : Flasks, scalpels)	Autoclave > Hot air oven
All culture media (Cm)	Autoclave
Exceptions in Cm : - Serum Cm (Loeffler's serum slope) - Egg Cm (Lowenstein - Jensen medium)	Inspissation > Tyndallization
Oily & powdery materials (E.g. : Liquid paraffin, glove dust powder)	Hot air oven

Disinfection

00:30:57

Types :Tests for Disinfectant :Efficacy tests :materials Disinfected :

- Flexible endoscopes :
Aldehyde (Glutaraldehyde > Orthophthalaldehyde) > H_2O_2 > ETO.
- Blood spill : 1% Na hypochlorite.
- Hand rub : Alcohol.
- Stethoscope/Thermometer :
Alcohol/Benzalkonium chloride.
Rigid endoscopes : Autoclave > H_2O_2 / ETO

Spaulding's Classification of medical Devices :

	medical devices	Level of disinfectant
Critical	- Enters sterile site - E.g. : Surgical instruments, implants	Sterilisation > HLD
Semi-critical	- Contact with mucous membrane - E.g. : Endoscopes	HLD
Non-critical	- Contact with intact skin - E.g. : Stethoscopes, sphygmomanometers	ILD/LLD

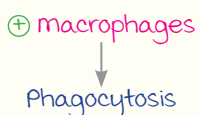
IMMUNOLOGY

Immunity

00:01:35

Defense system of the body.

Types :

	Innate immune system	Acquired immune system
Response	• minor • Present since birth	• major • Acquired after entry of organism
Recognition of organism	Genetic pattern recognised by : PRR (Pattern recognition receptors)	Antigens recognised by : APC (Antigen presenting cells)
Time taken	• Immediate action (3 - 4 hrs) • 1st line defense mechanism	• 5 - 7 days • 2nd line defense mechanism
MOA		• + T cells → Cell mediated immunity (CMI) • + B cells → Humoral immunity (Ab mediated)
Characteristics	• Non-specific • No memory	• Specific • memory cells formed

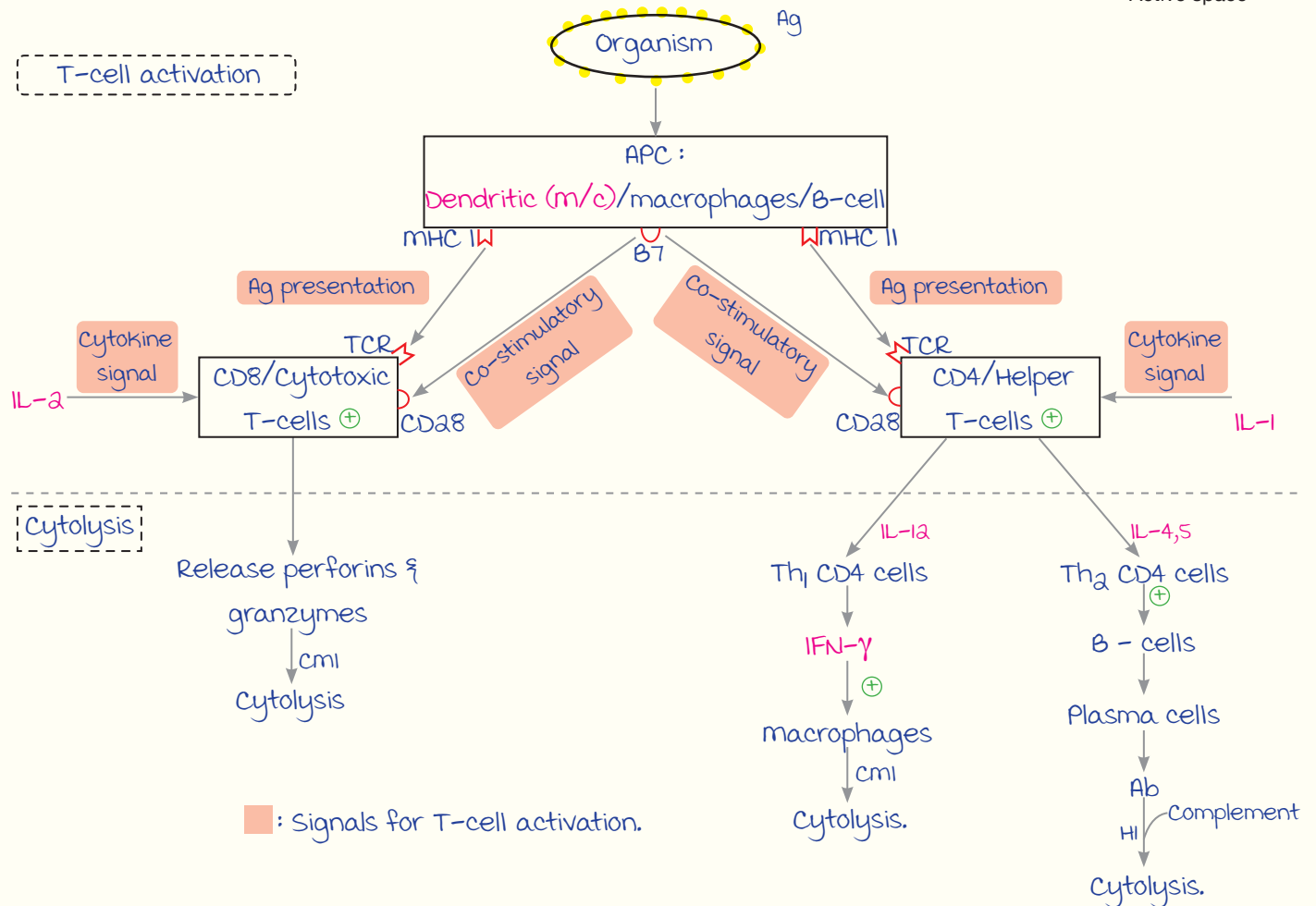
Acquired Immunity :

Types :

	Active acquired immunity	Passive acquired immunity
MOA	Antigen entry (into host)  Antibody production	Antibodies directly received by the host
Examples	• Infection • Vaccination	• IgG from mother to baby • IgG as post exposure prophylaxis • IgG via blood transfusion

mechanism :

----- Active space -----



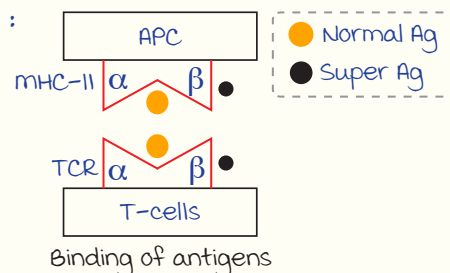
Note :

MHC gene : Chromosome 6p.

Antigen

00:17:55

Superantigen

Binds to lateral side of β -chain of the receptors :

Examples :

	TSST/pyrogenic exotoxin	Enterotoxin
Produced by	<i>S. aureus</i> > <i>S. pyogenes</i> (Group A streptococcus)	<i>S. aureus</i>
Leads to	Toxic shock syndrome	Gastroenteritis

----- Active space -----

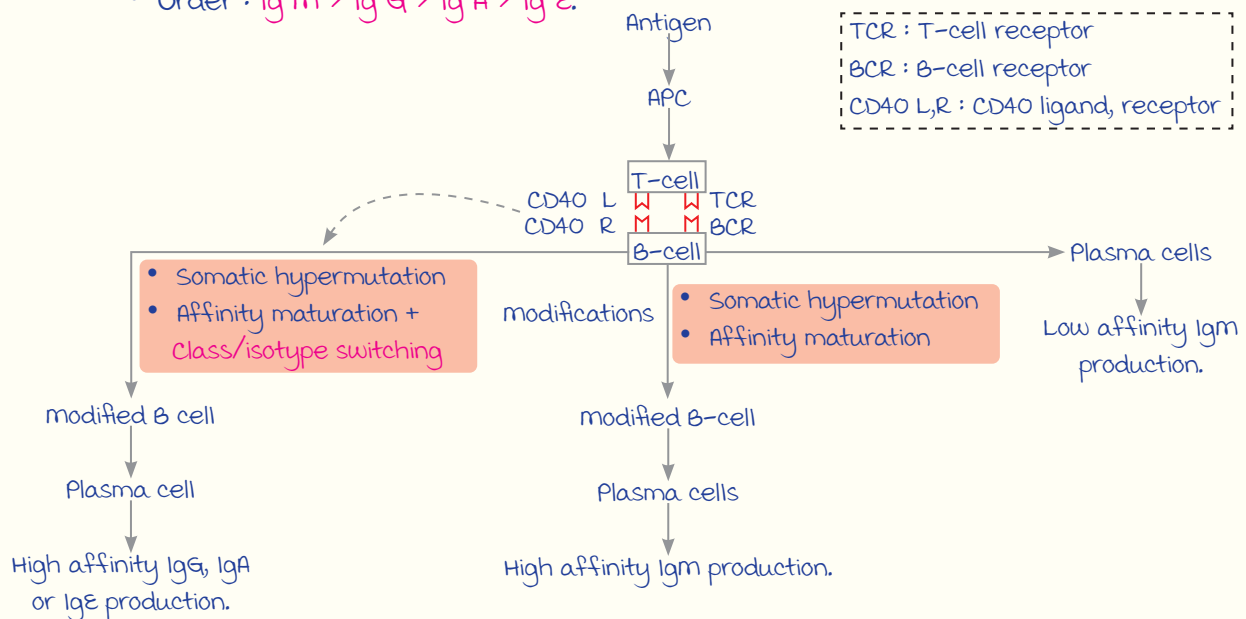
Antibody

00:21:47

- **Proteins** : Produced by **plasma** cells.
- **Detectable** : > **1 week** of symptoms (varies among organisms).

Formation 📌 :

- **m/c site** : **Lymph node**.
- **B-cell modification** : **Germinal centre** of LN.
- **Time taken** : 5 - 7 days after entry of organism.
- **Order** : **Ig M > Ig G > Ig A > Ig E**.

**Types** 📌 :

	Structure	Function	Key features
IgM	Pentamer	Fixes complement (Acquired immunity)	Highest molecular weight
IgG	monomer	• Fixes complement • Cross placenta ↓ Immunity to the baby	• Persists for years • Highest serum concentration • Subclass : - IgG1 : m/c - IgG2 : m/c against capsulated organisms - IgG3 - IgG4
IgA	Serum IgA	monomer	Fixes complement (Acquired immunity)
	Secretory IgA	Dimer	Local immunity : RT, GIT, GUT
IgD	monomer	-	Present in blood
IgE	monomer	-	Present in respiratory tract, GIT, GUT, breast milk
			Present as receptor on B cells (Along with IgM)
			↑ in : • Type I hypersensitivity • Parasitic infections

Hypersensitivity Reactions

00:33:17

----- Active space -----

Types :

	Type 1/allergic	Type 2/cytotoxic	Type 3/immune complex mediated	Type 4/delayed
	Allergy	RBC	Ag-Ab complex	Intracellular Ag
Examples	- Local : - Allergic rhinitis - Allergic bronchitis - ABPA - Systemic anaphylaxis	- Blood transfusion reaction - AIHA (Autoimmune hemolytic anemia) - Glomerulonephritis a/w Goodpasture syndrome	- Glomerulonephritis - Rheumatoid arthritis - Arthus reaction - Serum sickness - Type 2 lepra reaction	- Tuberculin/ mantoux test - Lepromin test - Type 1 lepra reaction - Allergic/contact dermatitis



 Exceptions

Transplant Immunology

00:40:24

HVGD vs GVHD :

	Host vs. graft disease (HVGD)/ Graft rejection	Graft vs. host disease (GVHD)
Definition	Host rejects grafts	Graft rejects host
Seen with	Any allograft	- Bone marrow allograft - Thymus allograft
Immune response	- Hyperacute (Hours) : HI (Type 2 Hypersensitivity) - Acute (months) : CMI (Type 4) - Chronic (Years) : CMI (Type 4)	Cell mediated (Type 4)

Note :

Allograft : From humans (Excluding a monozygotic twin).

----- Active space -----

Immunodeficiency Disorders (IDD)

00:44:37

		Inheritance	Defect	Key features
Humoral IDD	Bruton's disease (X-linked agammaglobulinemia)	X-linked	BTK gene mutation ↓ Bruton's tyrosine kinase deficiency Pre-B → → B-cell	Absent immunoglobulins
Cellular IDD	DiGeorge syndrome	AD	Chr. 22q11 deletion ↓ Abnormal development of 3rd & 4th pharyngeal pouches ↓ CATCH defects	CATCH defects : • Conotruncal cardiac defect • Abnormal facies • Thymic aplasia • Cleft lip & palate • Hypocalcemia
Combined (Humoral + cellular) IDD	SCID	X-linked (m/c)	IL-R gene → IL-receptor mutation → deficiency	-
		AR	ADA gene → ADA mutation → deficiency	-
	Wiskott-Aldrich syndrome	X-linked	WAS gene → No WASP protein	-
	Ataxia telangiectasia	AR	ATM gene → Defect in DNA repair	-
Phagocytic IDD	Chronic granulomatous disease	X-linked	Phox gene → NADPH oxidase mutation → deficiency	• Infection + granulomas • ↑ risk of infection : Catalase positive organisms (S. aureus). • Tests +ve : - Nitroblue tetrazolium - Dihydrorhodamine
	Chediak-Higashi disease	AR	LYST gene → Defect in phagosome lysosome fusion	• Albinism : Skin & eyes • Peroxidase +ve giant granules ⊕ in leucocytes

Complement Deficiency Disorders (CDD)

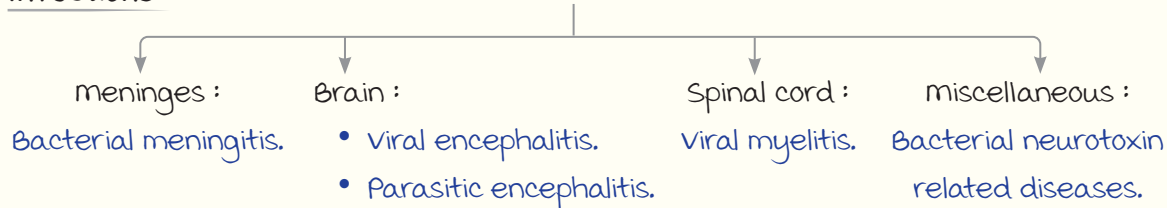
00:55:14

Complement/complement factors deficiency	Diseases
C1, C4, C2	Systemic lupus erythematosus
C3	Glomerulonephritis
C5b, C6, C7, C8, C9 (membrane attack complex)	Pyogenic disseminated infections : Neisseria infections
C _i esterase	Hereditary angioneurotic edema
DAF factor	Paroxysmal nocturnal hemoglobinuria
Factor I/H	Immune complex diseases

CENTRAL NERVOUS SYSTEM

----- Active space -----

Infections :



Bacterial Meningitis

00:02:05

Presentation :

Neck rigidity :
Difficulty in neck flexion.



Brudzinski's sign :

Neck flexion

Involuntary flexion of hips & knees.



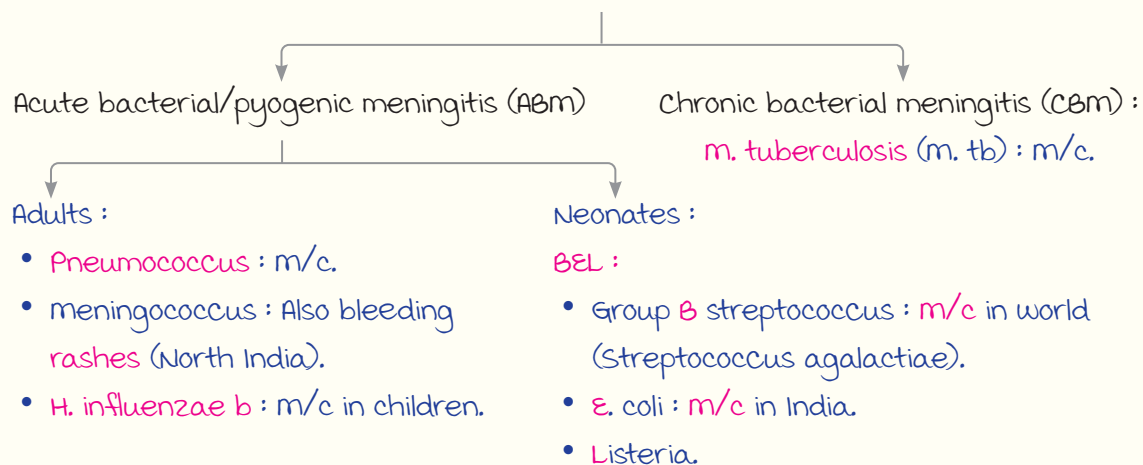
Kernig's sign :

Neck flexion

Resistance to knee extension
when hip is flexed.

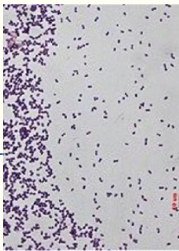
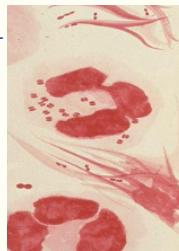
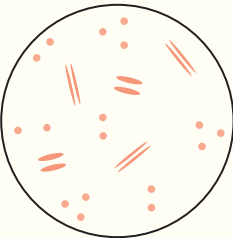
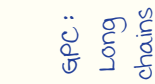
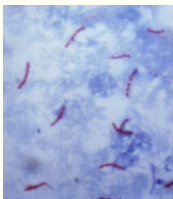
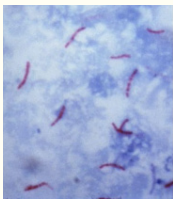
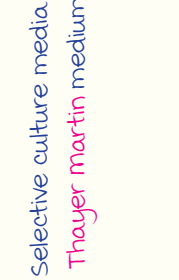
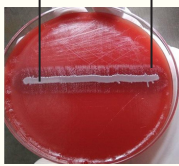


Organisms :



----- Active space -----

management 📌 :

	ABm (Adults)				ABm (Neonates)		CBm
CSF analysis	↑↑↑ Neutrophils & ↓↓↓ glucose						↑↑↑ Proteins : Cobweb formation
Organisms	Pneumococcus	meningococcus	Hib	GBS	E. coli	Listeria	mycobacterium tuberculosis
microscopy	- GPC : In pairs - Lens shaped, encapsulated 	- GPC : in pairs - Lens shaped, Encapsulated - Intracellular (Neutrophils) 	GN pleomorphic 	GPC : Long chains 	GNB 	GPB : Differential motility 22°C : Tumbling motility 37°C : Non-motile 	1 st step : - Acid fast - Non-flamentous 
Culture	Blood agar : α-hemolysis 	Selective culture media : Thayer martin medium 	Blood agar : Satellitism 	CAMP Test : +ve	-	-	Slow grower Culture not done
Biochemical tests	- Optochin sensitive - Bile soluble - Inulin fermenters	maltose fermenters	Non-specific			-	2 nd step : CBNAAT (PCR) : m. tb with rpo-B gene (Rifampicin resistance) 2 HRZE + 10 HRE - New : 2 HP2M + 10 HPm
Rx	Vancomycin + Ceftriaxone						Ampicillin + Ceftriaxone

GPC : Gram positive cocci GNB : Gram negative bacilli m : moxifloxacin
 GNC : Gram negative cocci P : Rifapentine

Viral Encephalitis

00:26:08

----- Active space -----

	Presentation	Vector	Lab Ix 	Prophylaxis	Rx
DNA viruses	- Seizures - Sporadic temporal lobe encephalitis (m/c)	-	PCR	-	Acyclovir
	- Seizures Progressive multifocal leukoencephalopathy (PML) - white brain disease - Seen in immunocompromised	-	PCR	-	-
RNA viruses	- Encephalitis : Seizures - Children (5 - 15yrs) - Northern India	Culex mosquito	Detect Ab	Vaccines : - Live : SA-14-14-a - Killed : Holar strain	-
	- Encephalitis : Seizures - West Bengal, Kerala	Fruit bat	PCR	-	-
	- Encephalitis : Seizures - Hyperactive/bizarre behaviour - Hydrophobia 100% mortality	Animal bite (m/c) : - World : Bat - India : Dog	- PCR : Saliva > CSF - Ag by DFA : Skin (Nape of neck) > Cornea	PEP : - Wash wound : Soap & water - Apply antiseptics - Immunoglobulins - Killed vaccine : Deltoid	-
	- Seizures - measles disease : Fever → Koplik spots → Rash ↓ 7 - 10 yrs (Reactivation)	-	Detect Ab	Live vaccine : Edmonston-Zagreb strain	-
	- SSPE : - School performance ↓ - myoclonus				

HSV : Herpes simplex virus SSPE : Subacute sclerosing panencephalitis
 JC : John Cunningham DFA : Direct fluorescent Ab test

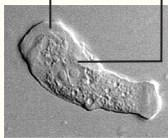
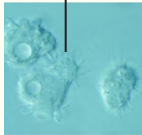
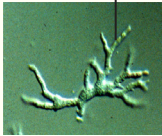
----- Active space -----

Parasitic Encephalitis

00:41:37

Transmission : Soil & water (Swimming).

Organisms : Free-living amoeba (Pseudopodia ⊕).

	Presentation	microscopy 		Rx
		Form	morphology	
Naegleria (m/c)	- Acute PAM - Seizures	Trophozoite	Lobe-like pseudopodia 	Liposomal amphotericin B
Sappinia	- Subacute SAE - Seizures	Trophozoite	Normal pseudopodia Bilobed nucleus 	-
Acanthamoeba	- Chronic GAE - Contact lens keratitis - Seizures	Trophozoite	Spike/thorn like pseudopodia 	-
Balamuthia	- Chronic GAE - Seizures	Trophozoite	Branching pseudopodia 	-

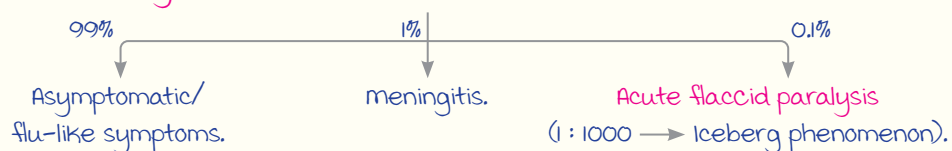
PAN : Primary Amoebic Meningoencephalitis
 SAE : Sappinia amoebic encephalitis
 GRAE : Granulomatous amoebic encephalitis

Viral Myelitis

00:51:37

Polio Virus :

Presentation : Poliomyelitis.



Ix : PCR (Stool).

Rx : Supportive.

Vaccine :

----- Active space -----

	OPV	IPV
Type	Live	Killed
Strain	Sabin	Salk
Schedule	Birth, 6, 10, 14 weeks 16 - 24 months	6, 14 weeks 9 months
Complications	- VDPV : Type 2 vaccine - VAPP : Type 3 vaccine	-

VDPV : vaccine derived polio virus

VAPP : vaccine associated paralytic poliomyelitis

Bacterial Neurotoxin Related Diseases

00:56:11

	Botulism	Tetanus
Organism	<i>Clostridium botulinum</i>	<i>Clostridium tetani</i>
Toxin	Botulinum toxin	Tetanospasmin
Pathogenesis	↓ Excitatory neurotransmitter : Ach ↓ muscle relaxation	↓ Inhibitory neurotransmitter : Glycine & GABA ↓ muscle contraction
Presentation	Flaccid paralysis	Spastic paralysis
Ix	Culture	Robertson cooked media
	Tests	Toxin : ELISA
Rx	Antitoxin + metronidazole/penicillin	

Tetanus vaccine :

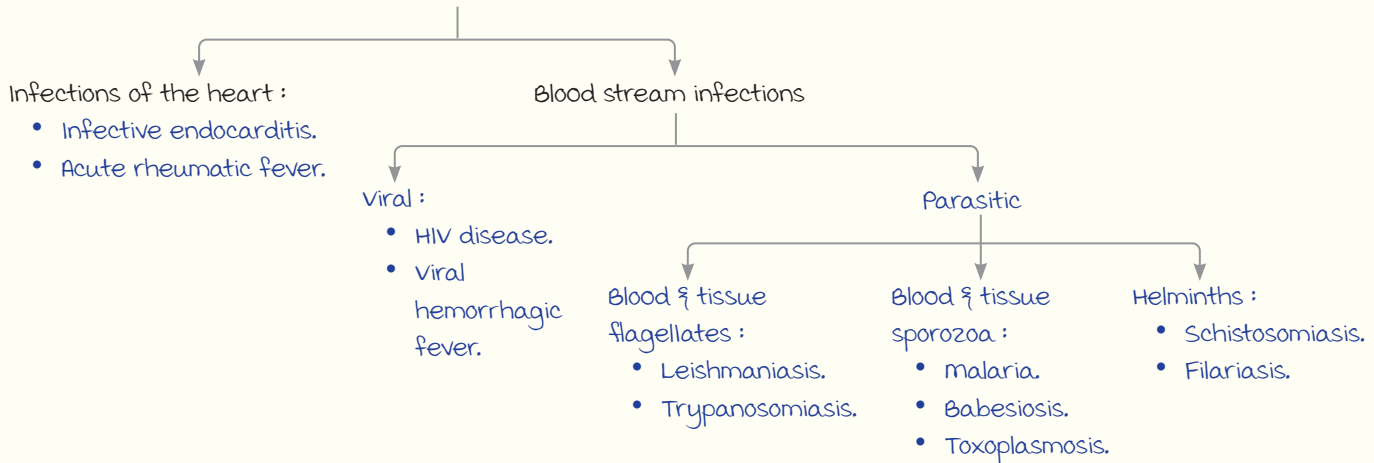
PEP protocol :

		Previous TT doses	
		≥ 3 doses	< 3 doses/unknown
Type of wound	Clean & minor	If last dose ≥ 10 years	Complete Td schedule
	Dirty & major	If last dose ≥ 5 years	Complete Td schedule + Immunoglobulins

----- Active space -----

CARDIOVASCULAR SYSTEM : INFECTIONS OF HEART

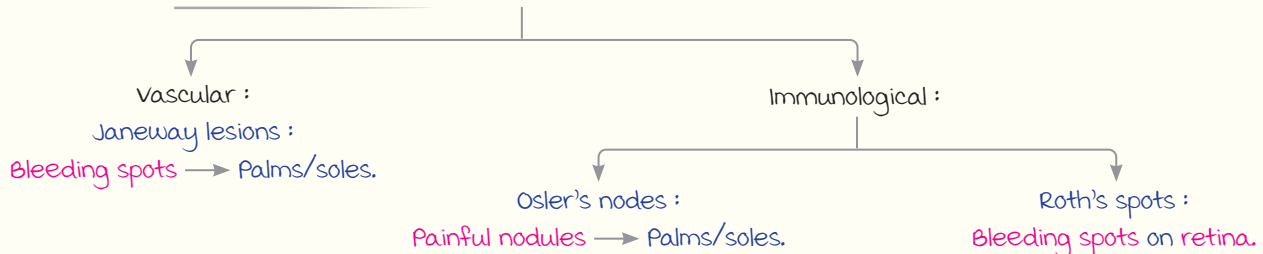
Diseases of CVS :



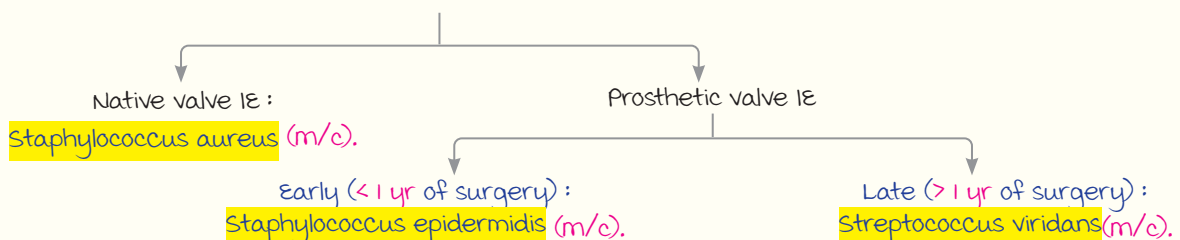
Infective Endocarditis

00:02:01

Clinical Features :



Etiology :



Diagnosis :

----- Active space -----

modified Duke's criteria (Updated) :

major criteria (3)		
microbiological criteria	Positive blood culture	• Typical organism : 2 separate +ve blood cultures • Atypical organism : ≥ 3 separate +ve blood cultures • Coxiella : Single +ve blood culture
	Positive lab Ix	Coxiella, Bartonella $\rightarrow$ PCR/IgG titre $\geq 1 : 800$
Imaging criteria	ECHO/CT	valvular vegetation/new valvular regurgitation/dehiscence
	FDG PET/CT	Abnormal metabolic activity in valve > 3 months
Surgical criteria	-	Direct visual evidence of IE during heart surgery

minor criteria (7)	
Fever	$> 38^{\circ}\text{C}$
Predisposing factors	Prior IE, prosthetic valve, CHD
vascular phenomenon	Janeway lesions, intracranial/conjunctival hemorrhage
Immunological phenomenon	Osler's nodes, Roth's spots
microbiological criteria	+ve blood culture not meeting major criteria
Imaging criteria	FDG PET/CT $\rightarrow$ Abnormal metabolic activity in valve < 3 months
Auscultation	New valvular regurgitation murmur

Diagnosis : 2 major OR 1 major + 3 minor OR 5 minor criteria.

Treatment :

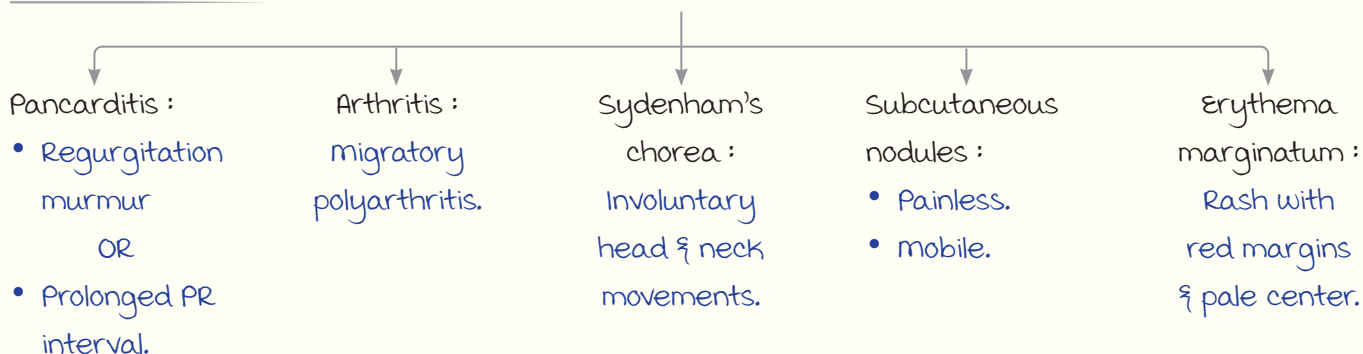
Native valve IE : $\left\{ \begin{array}{l} \bullet \text{ Penicillin} \\ \bullet \text{ Gentamicin} \\ \bullet \text{ Rifampicin} \end{array} \right. \xrightarrow{\text{If allergic}} \text{Vancomycin}$ } Prosthetic valve IE : 6 weeks.

4 - 6 weeks

Acute Rheumatic Fever (ARF)

00:17:08

Non-suppurative complication of group A streptococcal pharyngitis.

Clinical Features :

----- Active space -----

Diagnosis :

modified Jones criteria :

High risk	Both low & high risk	Low risk
monoarthritis ←	• Pancarditis • Arthritis • Sydenham's chorea • Subcutaneous nodules • Erythema marginatum	→ Polyarthritis
≥ 38°C ← monoarthralgia ← ≥ 30 mm/hr ←	• Fever • Arthralgia • ESR • CRP ≥ 3 mg/dL • Prolonged PR interval	→ ≥ 38.5°C → Polyarthralgia → ≥ 60 mm/hr

● major criteria (5)

● minor criteria (5)

Diagnosis : 2 major OR 1 major + 2 minor OR 3 minor + past h/o ARF.

Treatment :

Penicillin If allergic → Erythromycin.

CARDIOVASCULAR SYSTEM : BLOODSTREAM AND LYMPHATIC INFECTIONS

----- Active space -----

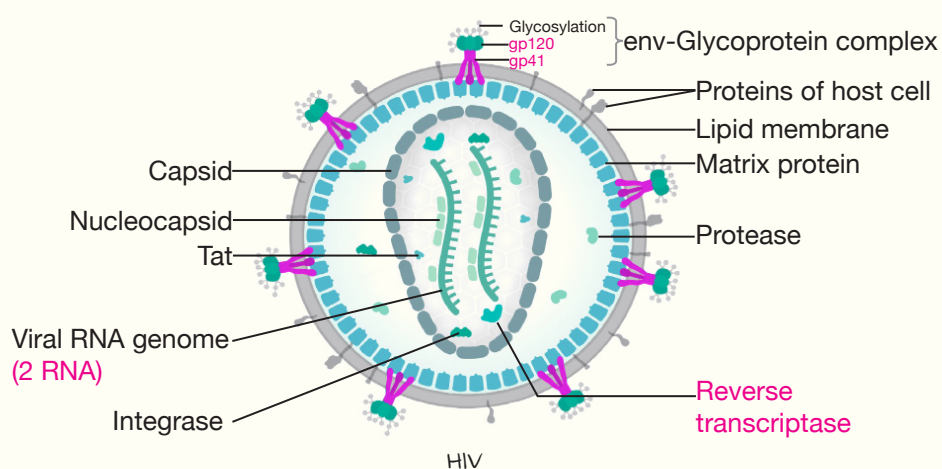
HIV

00:00:39

Retrovirus.

Structural genes :

1. **env** → env proteins : gp120, gp41.
2. **gag** → core proteins : p24 Ag.
3. **pol** → Polymerase, reverse transcriptase & integrase.



WHO Staging :

Stages	Features
Stage 1	- Asymptomatic - Generalised lymphadenopathy
Stage 2	- Oral aphthous ulcers - Herpes zoster/shingles
Stage 3	- White patches in oral cavity - On rubbing → Removed → Oral candidiasis (Candida) - Not removed → Oral hairy leukoplakia (EBV) - Pulmonary TB - Fever - Diarrhoea > 1 month
Stage 4 (AIDS)	- Oesophageal candidiasis - Wasting syndrome - Extrapulmonary TB - Cancers : Kaposi's sarcoma (m/c) - Other opportunistic infections

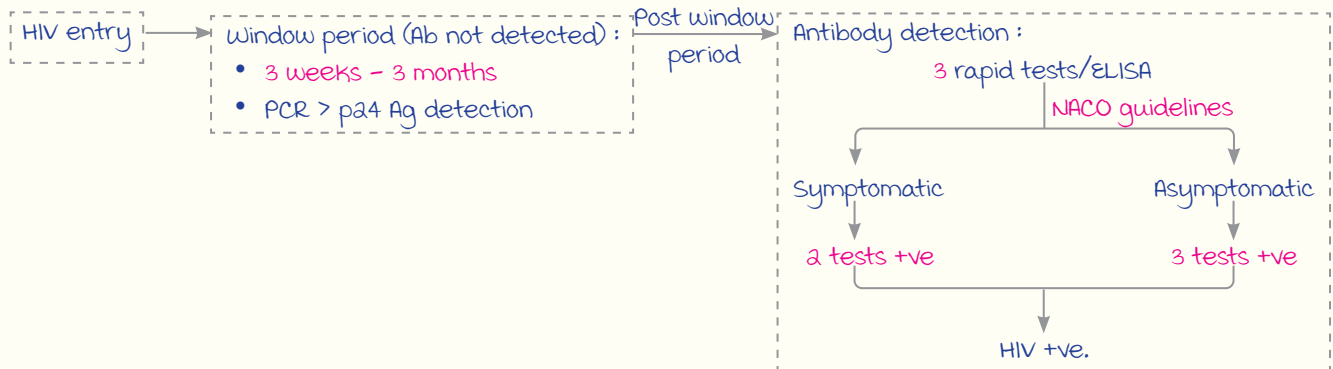
----- Active space -----

Opportunistic Infections Based on CD4 Count 🚨 :

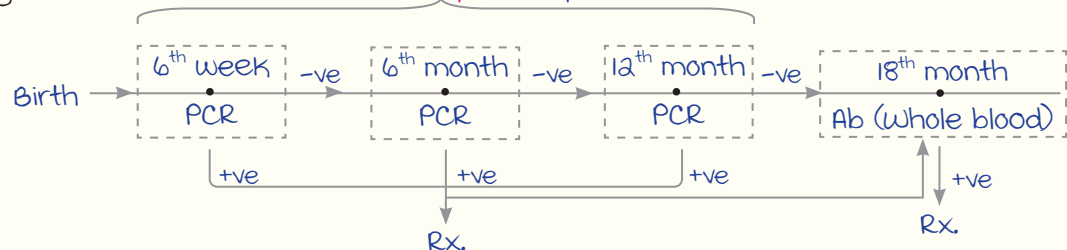
CD4 count/mm ³	Opportunistic infection	B : Bacteria F : Fungus V : Virus P : Parasite
< 500	- B : Pulmonary TB > Bacterial pneumonia (Pneumococcus) - F : Oral candidiasis - V : Herpes zoster (VZV)	
< 200	- P : Cryptosporidium diarrhoea - F : Pneumocystis pneumonia	
< 100	- P : Toxoplasmosis - F : Histoplasmosis	
< 50	- B : MAC (Mycobacterium avium-intracellulare complex) - F : Cryptococcosis - V : CMV infection	

Diagnosis :

Adult : Whole blood sample.



Congenital HIV infection : Dried blood spot sample



- PCR -ve but baby develops symptoms → Repeat PCR.

Treatment :

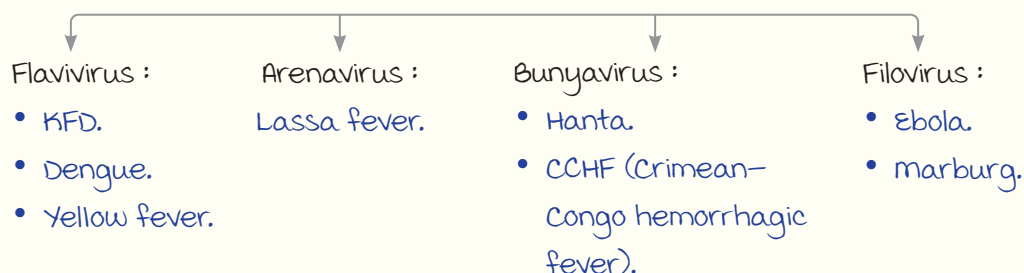
- HIV : TLD (Lifelong).
- Post-exposure prophylaxis : TLD (28 days).
- Pre-exposure prophylaxis (In high risk individuals) :
TE > TL (20 days).

T : Tenofovir
L : Lamivudine
D : Dolutegravir
E : Emtricitabine

----- Active space -----

Viral Hemorrhagic Fever (VHF)

00:18:05

Etiology :Flavivirus 🦠 :

	Kyasanur forest disease virus	Dengue virus	Yellow fever virus
vector	Ixodid tick (Hard tick)	Aedes aegypti > Aedes albopictus	
Reservoir	Rat/squirrel	-	
Amplifier	monkey		
Features	• Southern India (Karnataka) • VHF	• Primary infection : mild ↓ Classical dengue fever • Re-infection : Severe d/t ADE (Antibody dependent enhancement) ↓ - Dengue hemorrhagic fever - Dengue shock syndrome (DSS)	• Africa (⊖ in India). • VHF + hepatitis
Ix	PCR	Symptoms → < 5 days : PCR/NSI Ag → ≥ 5 days : Ab detection	PCR
Treatment	Supportive		
Vaccine	Killed vaccine	Live vaccine	Live vaccine (17-D strain)

----- Active space -----

Other viruses Causing VHF :

	Lassa fever	Hanta	CCHF	Ebola & marburg
vector	-		Ixodid tick	-
Reservoir	-			Fruit bat
Features	VHF + Pneumonia			VHF + Septic shock
Area	⊖ in India		Gujarat	⊖ in India
Investigations	PCR			
Treatment	Supportive			
Vaccine	⊖			

Blood and Tissue Flagellates

00:30:19

Leishmaniasis :

m/c seen in Bihar (Endemic).

Agent : *Leishmania* → Has kinetoplast & rk39 Ag.vector : Sand fly (*Phlebotomus*).

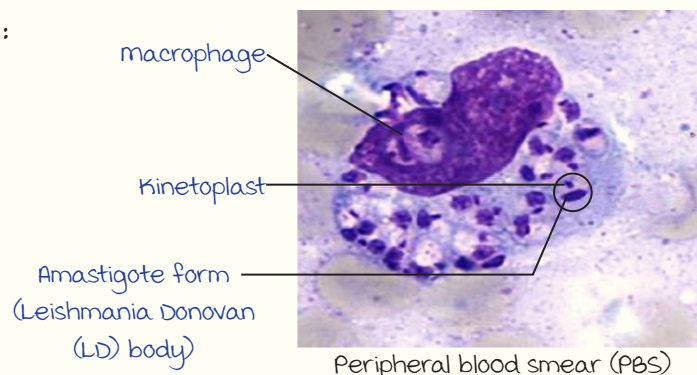
Presentations :

	Cutaneous/oriental sore/ Delhi boil/Baghdad button	mucocutaneous/espundia	visceral/kala-azar
Species	<i>L. tropica</i>	<i>L. braziliensis</i>	<i>L. donovani</i>
Clinical features	Papule	Ulcerative lesions	Hepatosplenomegaly ↓ Post kala-azar dermal leishmaniasis (Nodules on face)

Treatment :

- Liposomal Amphotericin B (iv).
 - miltefosine (Oral).
- } DOC.

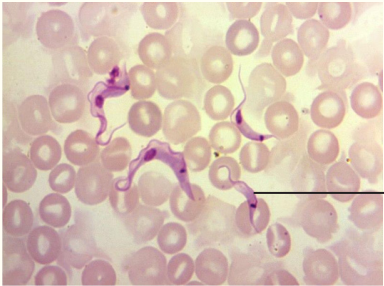
Investigations :



Trypanosomiasis :

----- Active space -----

Presentations :

	American	African
Disease	Chagas disease	Sleeping sickness
Agent	Trypanosoma cruzi	Trypanosoma brucei
vector	Reduviid bug	Tse-tse fly
Features	• mega : Esophagus, heart, colon • Romana's sign : u/L eyelid edema	• Somnolence • Winterbottom sign : Posterior triangle cervical lymphadenopathy
Peripheral blood smear	 Trypomastigote form Spindle shape with folded flagella	
Treatment	Benznidazole	Suramin

Blood and Tissue Sporozoa

00:40:41

malaria :

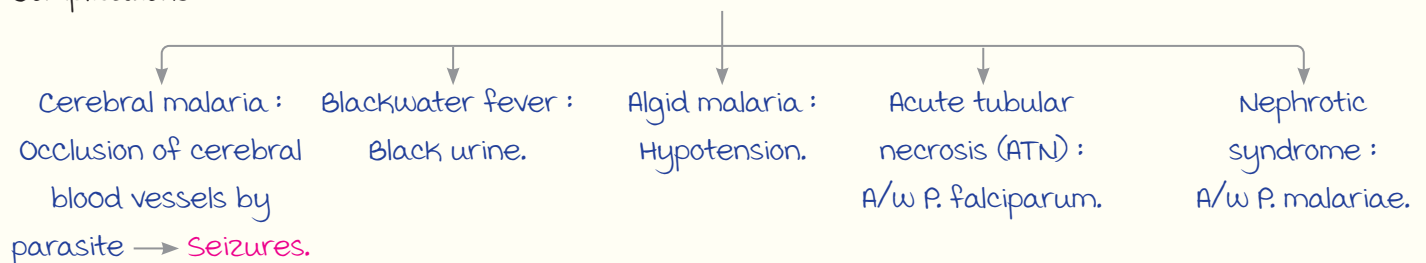
Agent : Plasmodium → P. falciparum > P. vivax.

vector : Female Anopheles mosquito.

Clinical features :

- Fever.
- Hepatosplenomegaly.
- Chills.

Complications :

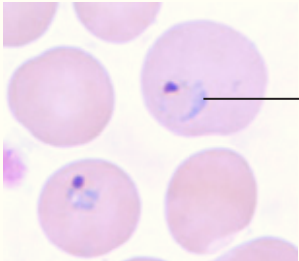
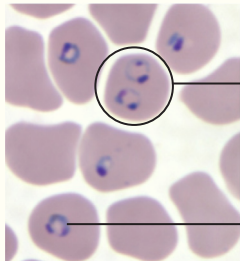
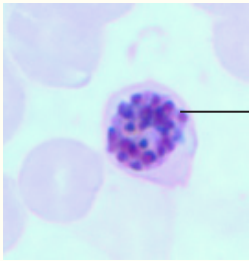
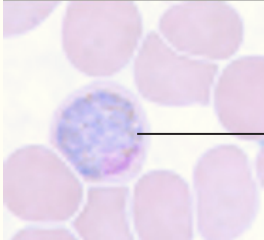
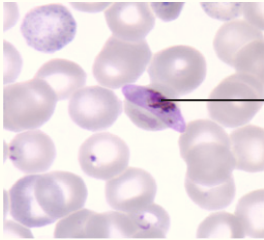
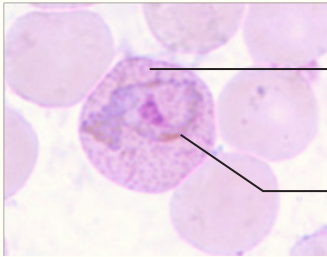
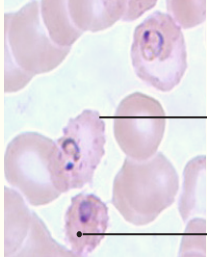


----- Active space -----

Investigations :

PBS :

- Thin smear for species identification.
- Giemsa/Leishman stain.

	P. vivax	P. falciparum
Antigen	pLDH	pLDH + HRP-2
Trophozoite	 Single ring form	 multiple ring forms Single ring forms
Schizont	 Round with merozoites	Present but not seen in PBS (Attached to capillary endothelium)
Gametocyte	 Round	 Banana/sickle shape
Granules & pigment	 Schuffner's dots (Smooth, fine) Brown pigment	 Maurer's cleft (Rough, bigger) Black pigment

pLDH : Plasmodium LDH

HRP-2 : Histidine rich protein-2

Treatment :

----- Active space -----

- *P. vivax* :
 - Chloroquine + Primaquine (Prevents relapse).
 - Pregnancy : Chloroquine.
- *P. falciparum* (Drug resistant) :

- Artesunate.
- Pyrimethamine + sulfadoxine.
- Primaquine (Kills gametocytes).

North-east India :

- Artemether.
- Lumefantrine.
- Primaquine.

Pregnancy :

- 1st tri : Quinine
- 2nd/3rd tri : Artesunate.

Babesiosis :

Agent : *Babesia*.

Vector : Ixodid tick.

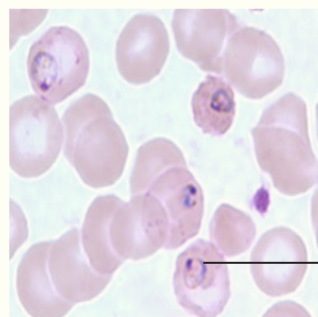
Clinical features :

- Fever.
- Chills.
- myalgia.

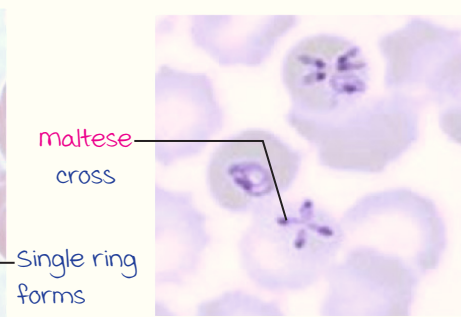
Treatment :

Atovaquone ± Azithromycin.

Investigations :



Trophozoite



merozoite

maltese cross

Single ring forms

Toxoplasmosis :

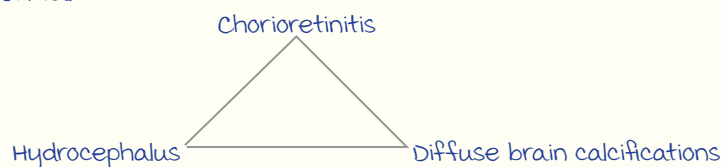
Agent : *Toxoplasma*.

Transmission :

Cat → Oocyst in stool → Sporulation → sporulated oocyst in soil (infective form) → Feco-oral route → Human.

Clinical features :

- Fever.
- Cervical lymphadenopathy.
- Immunocompromised :
 - myalgia (m/c)/Retinitis (Blurred vision)/Encephalitis (Seizures).
- Congenital :



----- Active space -----

Investigations 🦋 :

Immunological :

- Sabin-Feldman dye test (Specific) : Detects IgG.

- IgG avidity test :

Low avidity

High avidity

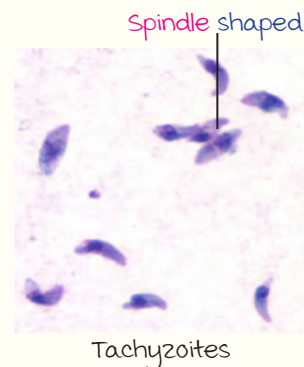
Acute infection.

Chronic infection.

Treatment :

- Pyrimethamine + Sulfadiazine
- Pregnancy : Spiramycin

} DOC.



Helminths

01:00:39

Schistosomiasis/Katayama Fever/Swimmer's Itch :

Agent : Schistosoma/blood fluke (Oviparous).

	S. hematobium	S. mansoni	S. japonicum
Features	Blood stream infection		
	Cystitis → Hematuria	Gastroenteritis → Diarrhea	
Cancer	Bladder cancer	No cancer	Colorectal cancer
Sample	Urine	Stool	
microscopy	 Terminal spine	 Lateral spine	 Lateral knob
Treatment	Praziquantel		

Filariasis :

North-east India (m/c).

vector : Culex mosquito.

Agents :

- Wuchereria bancrofti
- Brugia malayi

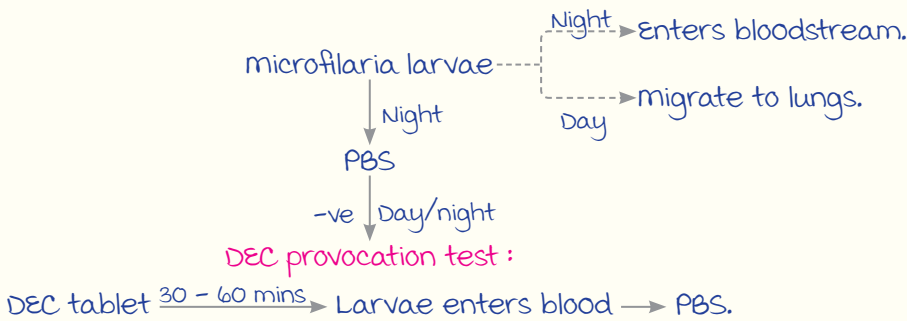
Lymphatic filariasis :

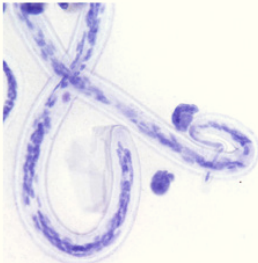
Obstruction → Elephantiasis.

- Loa loa : Calabar swelling
- Onchocerca volvulus : River blindness

Subcutaneous filariasis.

mx :

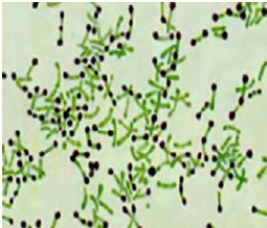
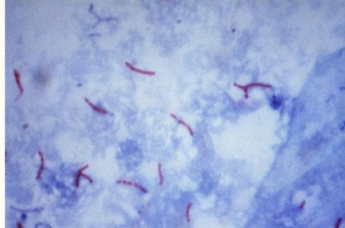


	w. bancrofti	B. malayi	Loa loa	Onchocerca
microscopy				
Sheath	⊕			⊖
Tail tip nuclei	⊖	2 nuclei	multiple	-
Treatment	DEC			Ivermectin

RESPIRATORY SYSTEM

Diphtheria, Pertussis and TB

00:01:56

	Diphtheria	Pertussis	Tuberculosis
Agent	<i>Corynebacterium diphtheriae</i>	<i>Bordetella pertussis</i>	<i>Mycobacterium tuberculosis</i>
Presentation	- Tonsillar area : Pseudomembrane - Bull neck lymphadenopathy - Complication : myocarditis	whooping cough	- Pulmonary TB : - Evening rise of temperature - Cough > 2 weeks - Weight loss - Extrapulmonary TB : - Lymphadenopathy (m/c) - meningitis : Neck rigidity - Spine : Back pain
microscopy	 Albert stain : metachromatic/volutin/ Babes-Ernst granules : Chinese letter pattern GS : Gram +ve bacilli	 Gram stain : Thumbprint appearance Gram -ve coccobacilli	 Acid fast stain : Acid fast Non filamentous GS : Gram +ve bacilli
Culture	- Potassium tellurite BA : Best - Loeffler's serum slope : Early diagnosis	Bordet - Gengou medium	LJ media (Not done)
Tests	Elek's gel precipitation test : Toxin demonstration	Ab - ELISA	- CBNAAT (PCR) : m. tb & rpo-B gene - Mantoux test - IGRA Latent TB
Rx	- Anti-toxin - Penicillin, erythromycin	Azithromycin	$2HRZE + 4HRE$ $\downarrow$ $2HPZM + 2HPM$

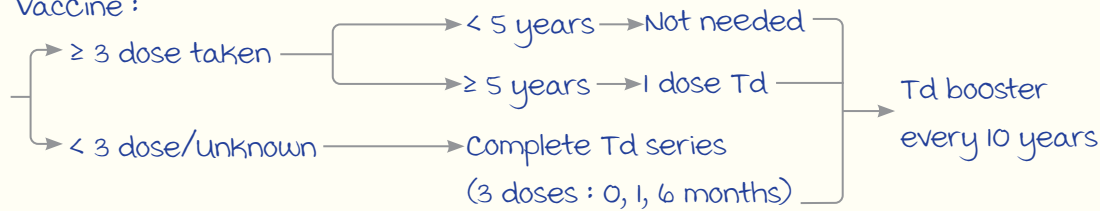
Changes in TB regimen :

- Rifampicin (R) → Rifapentine (P).
- Ethambutol (E) → moxifloxacin (M).
- 6 months → 4 months.

Diphtheria post-exposure prophylaxis :

- Antibiotic : Erythromycin/Penicillin x 10 days.

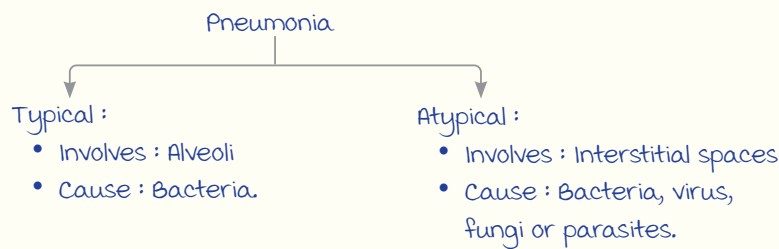
- Vaccine :



----- Active space -----

Typical/Lobar Pneumonia

00:20:00



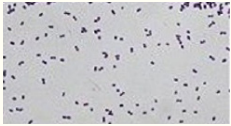
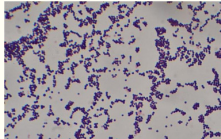
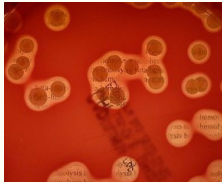
c/f : Cough, dyspnea, consolidation.

Organism :

Bacteria	m/c cause
Pneumococcus	Community AP
Staphylococcus	Hospital AP
Pseudomonas	In adults with cystic fibrosis

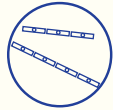
Bacteria	m/c cause
Acinetobacter	ventilator associated pneumonia (VAP)
Burkholderia	Zoonotic (Horse)
Bacillus anthracis	Zoonotic (Cattle)

Diagnosis and Rx :

	Pneumococcus	Staphylococcus aureus
Gram stain	 Gram +ve cocci : • In pairs/lens shaped • Capsulated	 Gram +ve cocci : • In clusters
Blood agar	 α hemolysis	 β hemolysis
Biochemical tests	• Optochin sensitive • Bile soluble • Inulin fermenter	Coagulase +ve
Rx	• mild : Amoxicillin • Severe : Ceftriaxone	• MSSA : Penicillin • MRSA : Vancomycin • VRSA : Daptomycin/linezolid

----- Active space -----

	Pseudomonas	Acinetobacter
Gram stain	Gram -ve bacilli	Gram -ve coccobacilli
Culture	Cetrimide agar	Blood agar
Rx	Ceftazidime/meropenem	

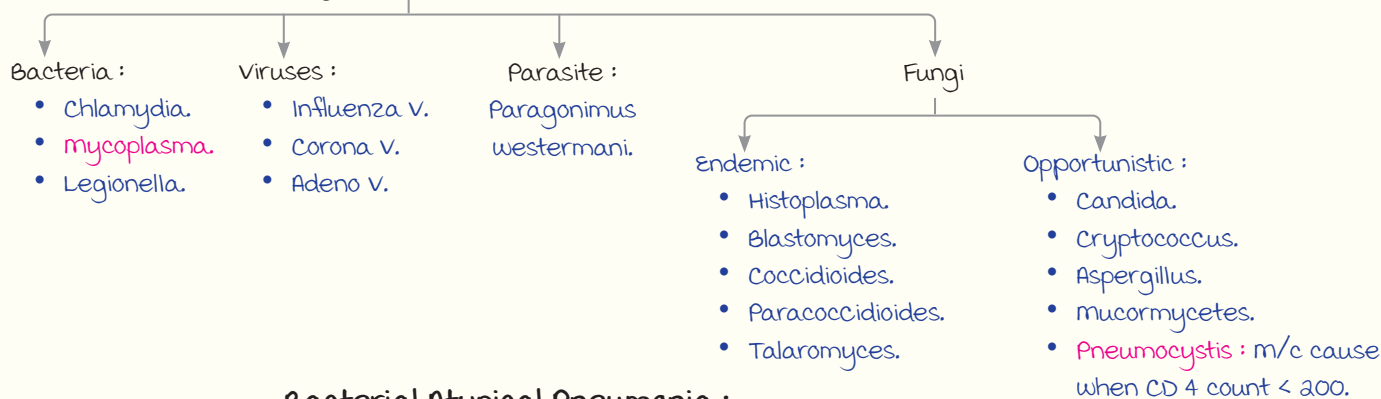
	Burkholderia	Bacillus anthracis
Disease	B. pseudomallei : melioidosis (Soil and water) B. mallei : Glander's disease (Horse) B. cepacia : Cepacia syndrome (Contaminated IVF)	Wool sorters disease : Hemorrhagic pneumonia
Gram stain	Gram -ve bacilli	- Gram +ve bacilli - In chains - Non-budding spores - Bamboo stick appearance 
Culture	Blood agar	Blood agar : - medusa head appearance - String of pearl appearance
Rx	Ceftazidime/meropenem	Ciprofloxacin/Doxycycline

Atypical/Interstitial Pneumonia : Bacterial, Viral, Parasitic

00:36:26

c/f : Cough, dyspnea.

Organisms :

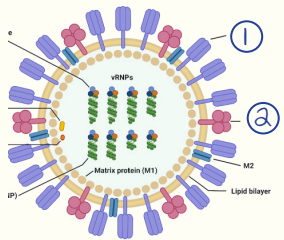
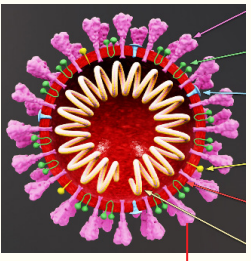
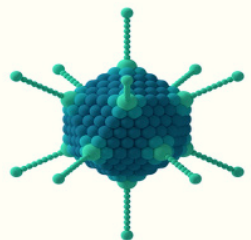


Bacterial Atypical Pneumonia :

	Chlamydia	mycoplasma	Legionella
Features	-	m/c cause of atypical pneumonia	Fresh water sources : Whirlpools, humidifiers
GS	Not seen		
Culture	Tissue culture : - McCoy cell culture - HeLa cell culture	PPLO agar : Fried egg colonies	BCYE agar
Diagnostic test	PCR		- Legionella urine Ag - PCR
Rx	Azithromycin > Levofloxacin		

Viral Atypical Pneumonia :

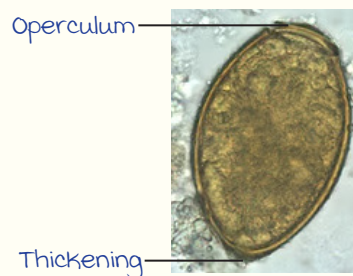
----- Active space -----

	Influenza A and B (m/c)	Corona	Adeno
Structure	 8 segmented nucleic acid	 Spike receptors	 Satellite shape
Receptors	① Hemagglutinin ② Neuraminidase	Spike/flower petal	Long antenna like
Outbreaks	• Pandemic : Ag shift (Genetic reassortment) • Epidemic : Ag drift (Point mutation)		-
Diagnosis	PCR		
Rx	Oseltamivir	-	
Vaccine	• Live : Inhalational • Killed : Im (Advised)	• Subunit (s. Ag) : Covishield • Killed : Covaxin	

Parasitic Atypical Pneumonia :

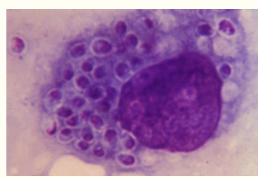
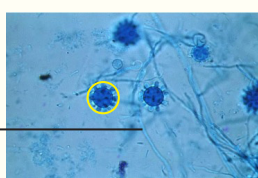
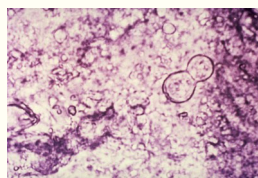
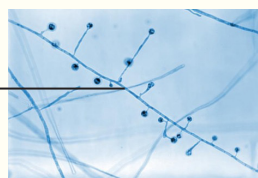
Paragonimus westermani :

- AKA Lung fluke.
- Oviparous : Egg in stool & sputum.
- Rx : Praziquantel.

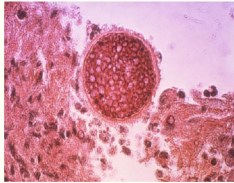
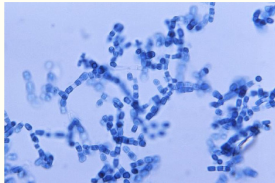
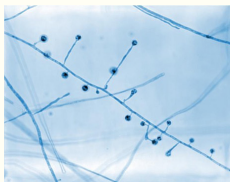
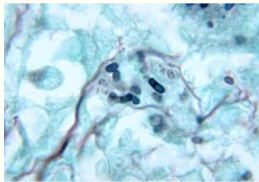
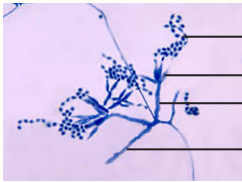
Fungal Atypical Pneumonia

00:53:30

Endemic (Dimorphic) :

Agent	Disease	Yeast	mould
Histoplasma	Histoplasmosis/ darling's disease/ Ohio valley fever	 Intracellular (macrophage)	 Hyphae Tuberculate conidia
Blastomyces	Blastomycosis/ Gilchrist's disease/ north american blastomycosis	 Broad based budding yeast	 Hyphae Pear shaped conidia

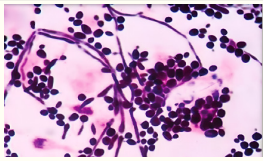
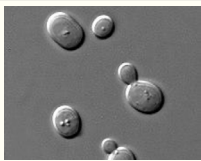
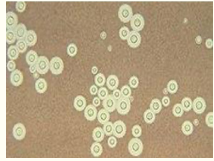
----- Active space -----

Agent	Disease	Yeast	mould
Coccidioides	Coccidioidomycosis/ valley fever/ Desert rheumatism	 Spherule with endospores	 Arthrospores Rectangular conidia
Para-coccidioides	Paracoccidioidomycosis/ South american blastomycosis	 mickey mouse appearance	 Pear shaped conidia
Talaromyces marneffei (SDA : Red pigment)	Talaromycosis/ Penicilliosis / South east asia	 Oval yeast with septum	 Bottle brush appearance Conidia Phialides Conidiophore Hyphae

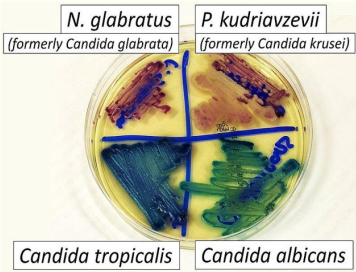
Rx :

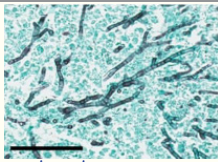
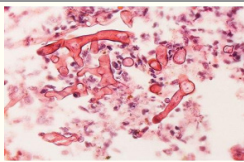
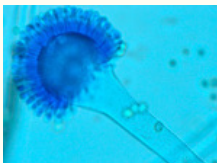
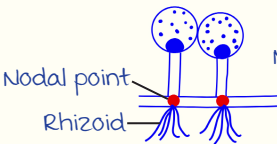
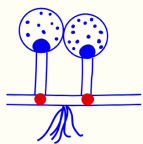
- IV liposomal amphotericin B.
- Oral itraconazole.

Opportunistic :

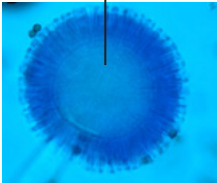
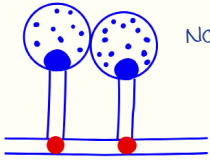
	Candidiasis	Cryptococcosis
Agent	Candida : Yeast like	Cryptococcus : Yeast
Disease	• Skin and mucosa : White membrane, removed on rubbing • Lung (Invasive) : Pneumonia	• Lung : Pneumonia (m/c) • CNS : meningitis
microscopy	 Budding yeast with pseudohyphae	 Budding yeast
Tests	 Germ tube test : (Reynolds - Braude phenomenon) • C. albicans (m/c) • C. dubliniensis } +ve	 Negative staining : Halo (Capsule)
Invasive marker	β - DG	

----- Active space -----

	Candidiasis	Cryptococcosis
Culture	 <i>N. glabratus</i> (formerly <i>Candida glabrata</i>) <i>P. kudriavzevii</i> (formerly <i>Candida krusei</i>) <i>Candida tropicalis</i> <i>Candida albicans</i> Chrome agar : Identify all species	 Bird seed agar : Brown colonies (melanin)
Rx	Oral Fluconazole ↓ Resistance Echinocandins Topical : Clotrimazole	Lip-amphotericin B + Flucytosine } 2 wks ↓ Fluconazole x 8 wks (Continue until : CD4 count > 100)

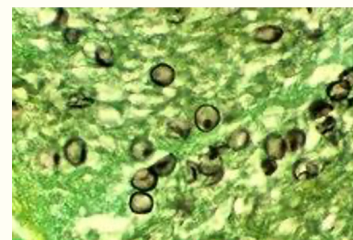
	Aspergillosis	mucormycosis/zygomycosis
Agent	Aspergillus : <i>A. fumigatus</i> : m/c <i>A. flavus</i> <i>A. niger</i>	mucormycetes : <i>Rhizopus</i> : m/c - Absidia - mucor
Risk factors	Neutropenia	- DKA - Iron therapy - Post covid
Disease	- Pneumonia (m/c) - Sinusitis	Rhinocerebral mucormycosis : m/c (Black lesions) - Pneumonia
Tests	β -DG and galactomannan : Invasive markers	mRI : head and neck
Culture	 - Narrow hyphae - Acute angle Dichotomous branching	 Broad/ribbon hyphae - Obtuse and right angle branching
LPCB stain	Surrounding conidia :  1/3rd vesicle : <i>A. fumigatus</i>  2/3rd vesicle : <i>A. flavus</i>	 Nodal point Rhizoid Nodal : <i>Rhizopus</i>  Internodal : <i>Absidia</i>

----- Active space -----

	Aspergillosis	mucormycosis/zygomycosis
LPCB stain	vesicle  Complete vesicle : A. Niger	 No rhizoid : mucor
Rx	voriconazole	Surgery ↓ Lipo-amp-B and posaconazole

Pneumocystosis :

- Agent : *Pneumocystis jirovecii*/carinii.
- Invasive marker : β -DG.
- Disease : Pneumonia.
- Sample : BAL > Sputum.
- Test : GMS > KOH wet mount.
- Rx : Cotrimoxazole.



GMS : Crushed ping pong ball appearance
(black cysts)

GASTROINTESTINAL & HEPATOBILIARY SYSTEM

----- Active space -----

Peptic Ulcer Disease

00:00:56

Causative organism : *Helicobacter pylori*.

Clinical presentation : Peptic ulcers → Gastric carcinoma.

Diagnosis :

microscopy : Gram -ve, comma shaped.

urease test : **Positive** (Differentiate from campylobacter).

motility : **Corkscrew**.

Culture media : **Skirrow's** media.

Treatment :

	Triple therapy	Quadruple therapy
Drugs	• Clarithromycin • metronidazole/amoxicillin • Omeprazole	• Tetracycline • metronidazole • Omeprazole • Bismuth sulphate
Duration	2 weeks	

Bacterial Gastroenteritis

00:04:12

Clinical features :

- Fever.
- Vomiting.
- Abdominal pain.
- **Diarrhoea/dysentery**.

Organisms Causing G.E. :

Organisms producing pre-formed toxins :

Organism	Incubation period	Toxin	Food products
<i>Staphylococcus</i>	< 6 hrs	Enterotoxin	milk/milk products
<i>Bacillus cereus</i>	< 6 hrs	Emetic toxin	Fried rice
<i>Clostridium botulinum</i>	> 16 hrs	C_a botulinum toxin	Canned food

Organisms producing post-formed toxins :

Incubation period : > 6 hrs.

Organism	Toxin	Food product	Symptoms
<i>Vibrio cholerae</i>	Cholera toxin	Sea food	Rice watery diarrhoea
EHEC (O157 : H7)	verocytotoxin (Shiga-like toxin)	milk & vegetables	Dysentery & HUS
<i>Shigella</i>	Shiga toxin	Egg & vegetables	Dysentery & HUS

----- Active space -----

Non toxin-producing organisms :

	V. Parahemolyticus	Campylobacter	Salmonella
Diseases caused	GE, wound infection	GE	Typhoid fever
microscopy	Gram -ve, comma shaped	Gram -ve, 'S' shaped	Gram -ve bacilli
Culture	- TCBS agar : Green colonies - Wagatsuma agar : Kanagawa phenomenon	Skirrow's media	Wilson Blair media
Lab. tests	- String test : Negative - Cholera red reaction : Negative	urease test : Negative	Widal test : Positive

Salmonella carriers :

- Source of infection.
- Types
 - urine carriers.
 - Stool carriers.
- Screening : Vi antibody test.
- Confirmatory : urine/stool culture.

Vibrio cholerae vs. parahemolyticus :

	Vibrio cholerae	Vibrio parahemolyticus
microscopy	Gram -ve, comma shaped	
	- Fish in stream appearance - Darting motility	-
TCBS agar	Yellow colonies	Green colonies
Wagatsuma agar	-	Kanagawa phenomenon
String test	+ve	-ve
Cholera red reaction	+ve	-ve
Treatment	Doxycycline	

Viral Gastroenteritis

00:15:23

Clinical features :

- Abdominal pain.
- Vomiting.
- Diarrhoea.

Rotavirus :

- dsRNA with 11 segments.
- Diagnosis : Antigen ELISA (Stool sample).
- vaccine : Live (Oral).

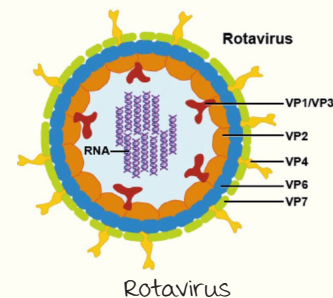
Causative Organisms :

Reo (m/c) : Rotavirus (m/c in < 2 yrs age).

Adeno : Adenovirus 40, 41.

Calici : Norovirus (m/c in adults) & sapovirus.

Astro : Astrovirus.



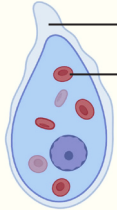
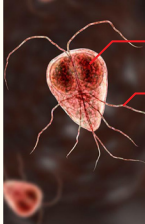
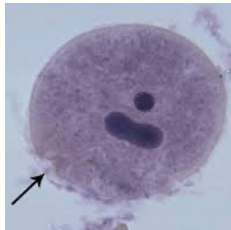
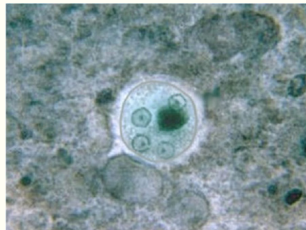
Parasitic Gastroenteritis

00:18:40

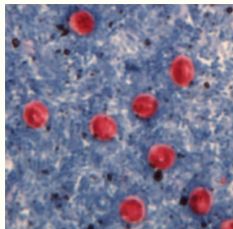
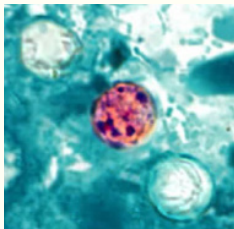
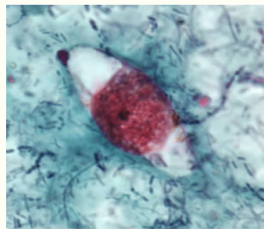
----- Active space -----

Protozoal Gastroenteritis :

Non acid fast protozoal gastroenteritis :

		Balantidium coli	Entamoeba histolytica	Giardia lamblia
Disease		Balantidiasis	Amoebiasis	Giardiasis
Clinical features		Dysentery	- Dysentery - Flask shaped ulcer	- Diarrhoea - Steatorrhoea
Lab diagnosis	Trophozoites (Locomotory organ)	 Cilia	 Pseudopodia RBC	 Suckers (2) Flagella (8) Falling leaf motility
	Cysts	 - Round shape 2 nuclei	 - Round shape 4 nuclei	 - Oval shape 4 nuclei
Treatment		Tetracycline > doxycycline	- metronidazole - For cyst carriers : - Diloxanide furoate - Paromomycin	metronidazole

Acid fast protozoal gastroenteritis :

	Cryptosporidium	Cyclospora	Cystoisospora
Clinical features	Diarrhoea		
Laboratory diagnosis : Cysts	 - Round shape 4 - 5 μ	 - Round shape 8 - 10 μ	 - Oval shape 15 - 30 μ
Treatment	Nitazoxanide	Cotrimoxazole	

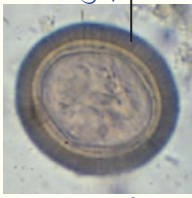
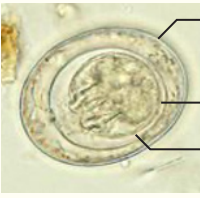
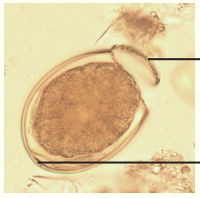
Cryptosporidium : Cause autoinfection.

----- Active space -----

Helminthic Gastroenteritis

00:27:35

Cestodes :

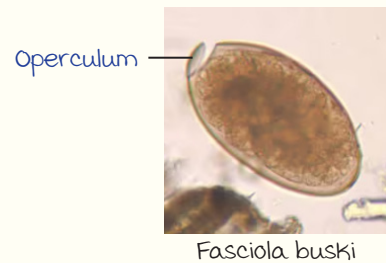
	Taenia solium	Taenia saginata	Hymenolepis nana	Diphyllobothrium latum
AKA	Pork tapeworm	Beef tapeworm	Dwarf tapeworm	Cyclops & fish tapeworm
Hosts	man & pig	man & cattle	Only man	man, cyclops & fish
Diseases	• GE (Taeniasis) • Neurocysticercosis : - c/f : Seizures - Rx : Steroids + albendazole	GE (Taeniasis)	GE	• GE • megaloblastic anemia (↓ Absorption of vit. B₁₂)
Lab diagnosis	Embryophore  Non acid fast	 Acid fast		
Treatment	Praziquantel			

All cestodes : Oviparous.

Trematode :

Fasciola buski (Intestinal fluke) :

- Hosts : man, snail & aquatic plants.
- Disease : GE.
- Lab diagnosis : Eggs in stool (Ovoparous).
- Rx : **Praziquantel.**

**Nematodes :**

- Oviparous (Exception : **Strongyloides** → Ovoviviparous).
- Host : Only man.

Treatment :

- Strongyloides : Ivermectin.
- Other nematodes : Albendazole.

Nematodes causing GE :

----- Active space -----

	Diseases	Laboratory diagnosis
<i>Strongyloides stercoralis</i> (Parthenogenic worm)	- Diarrhoea - Cutaneous larva migrans - Loeffler's pulmonary eosinophilia - Hyperinfection syndrome in immunocompromised (Opportunistic)	 Larva in stool
<i>Ancylostoma duodenale</i> / <i>Necator americanus</i> (Hookworm)	- Diarrhoea - Cutaneous larva migrans - Loeffler's pulmonary eosinophilia - Iron deficiency anemia	 Blastomeres inside embryo Inner membrane Outer membrane
<i>Ascaris lumbricoides</i> (Roundworm)	- Diarrhoea - Visceral larva migrans - Loeffler's pulmonary eosinophilia - Intestinal obstruction	 Semilunar chitin sheath (Both poles of embryo)
<i>Enterobius vermicularis</i> (Pinworm)	- Perianal itching - Appendicitis	 Planoconvex (D-shaped)
<i>Trichuris trichiura</i> (Whipworm)	- Chronic dysentery ↓ Iron deficiency anemia - Rectal prolapse	 mucus plugs (Both sides)

Viral Hepatitis

00:44:50

Hepatitis A & E :

	Hepatitis A	Hepatitis E
Transmission	Feco-oral	
Disease	Acute hepatitis	
Clinical features	Usually asymptomatic	High mortality in pregnancy
Lab diagnosis	IgM antibody ELISA	
Treatment & vaccine	Not available	

----- Active space -----

Hepatitis B, C & D :

	Hepatitis B	Hepatitis C	Hepatitis D
Route of transmission	• Parenteral • Sexual • Vertical		
Diseases	Acute hepatitis → Chronic hepatitis → Hepatocellular carcinoma		
Lab diagnosis	• HBsAg : +ve → Infection • Anti-HBc : ↳ IgM : Acute ↳ IgG : Chronic • HBeAg : - +ve : High infectivity - -ve : Low infectivity	• IgM HCV : 1st test • PCR : Confirmatory test	With HBV Coinfection : • IgM HDV ⊕ • IgM HBV ⊕ Superinfection : • IgM HDV ⊕ • IgG HBV ⊕
Treatment	Tenofovir > 1 yr	Sofosbuvir + velpatasvir	Not available
Vaccine	Subunit vaccine (HBsAg)	Not available	Not available

Parasitic Hepatitis

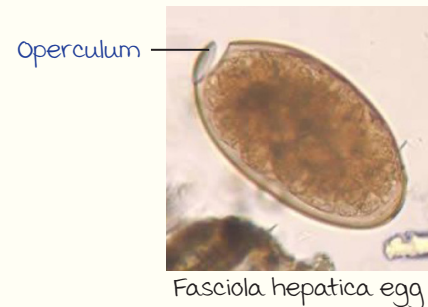
00:52:07

Liver Flukes :

Fasciola hepatica :

- Hosts : man, snail & aquatic plants.
- Diseases : Hepatitis & **cholangitis**.
- Laboratory diagnosis :
Identification of egg in stool (Oviparous).
- Treatment : Triclabendazole.

Opisthorchis viverrini } Hepatitis, cholangitis,
 Clonorchis sinensis } **cholangiocarcinoma**.

**Hydatid Cystic Disease**

00:54:00

Causative organism : Echinococcus granulosus (Dog tapeworm).

Hosts : man & dog.

Clinical manifestation : Formation of hydatid cyst in **liver (m/c)** > lung

unilocular cyst : E. granulosus.

multilocular/alveolar cyst : E. multilocularis.

Diagnosis :

- Imaging : USG/CT (Hydatid cyst showing **water lily sign**).
- Laboratory investigation : Antibody ELISA (Confirmatory).

Treatment :

- PAIR : Puncture aspiration injection reaspiration $\xrightarrow{\text{Fails}}$ Surgery.
- **Albendazole** (DOC).

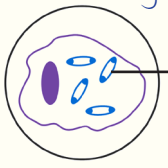
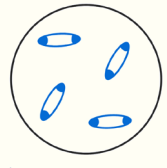
GENITOURINARY SYSTEM

----- Active space -----

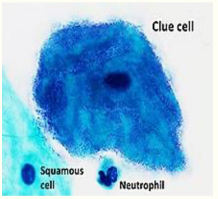
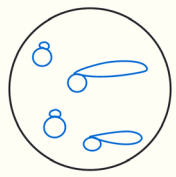
Diseases of Genital System

00:01:22

Genital Ulcerative Diseases :

	Granuloma inguinale/ Donovanosis	Chancroid	Syphilis	Lymphogranuloma venereum (LGV)
Etiology	<i>Klebsiella granulomatis</i>	<i>Haemophilus ducreyi</i>	<i>Treponema pallidum</i>	<i>Chlamydia trachomatis</i>
c/f	- Painless ulcer - Pseudo bubos	- Painful ulcer - Painful bubos	- Painless ulcer - Painless bubos	- Painless ulcer - Painful bubos
microscopy	Donovan body : 	 Safety pin appearance (Outside macrophage)	Dark field microscopy :  Spiral spirochetes in dark background	Not seen in Gram stain
Rx	Azithromycin		Penicillin	Azithromycin

Vaginitis :

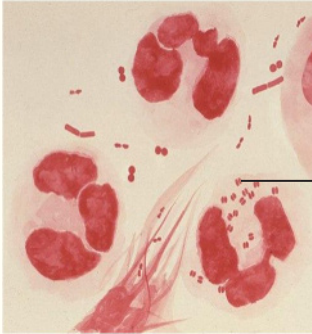
	Parasitic	Bacterial	Fungal
Etiology	<i>Trichomonas vaginalis</i>	<i>Gardnerella vaginalis</i>	<i>Candida albicans</i>
Itching	mild	None	Severe
vaginal discharge	Yellow-green, fishy smell	white, fishy smell	Curdy white
microscopy	Trophozooid : 5 flagella, 1 sucker - Jerky/twitching motility 	Clue cells : vaginal epithelial cells coated with <i>G. vaginalis</i> 	Budding yeast with pseudohyphae 
Rx	metronidazole		Fluconazole

----- Active space -----

Diseases of Urinary System

00:10:30

urethritis 📌 :

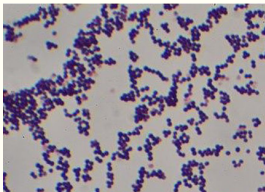
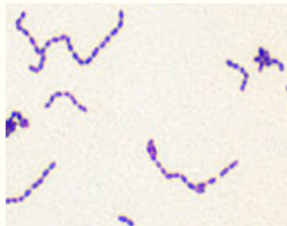
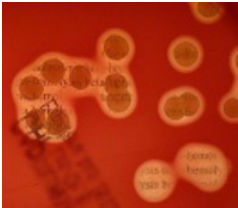
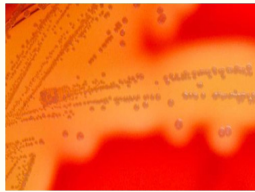
	Gonococcal urethritis/gonorrhoea	Non-gonococcal urethritis
Incubation period	< 1 week	> 1 week
Etiology	Gonococcus/Neisseria gonorrhoeae	• Bacteria → Chlamydia trachomatis (m/c) → Mycoplasma genitalium → Ureaplasma urealyticum • Virus : HSV-2 • Fungus : Candida albicans • Parasite : Trichomonas vaginalis
c/f	Purulent urethral discharge	Mucopurulent urethral discharge
microscopy	 GNC in pairs (kidney shaped) inside neutrophils	-
Culture	• Thayer-Martin medium • Non-maltose fermenter	-
Rx	Ceftriaxone	-

SKIN, SUBCUTANEOUS & MUSCULOSKELETAL SYSTEM

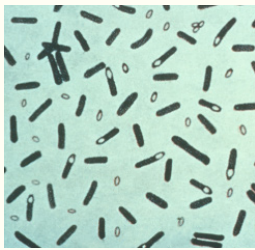
----- Active space -----

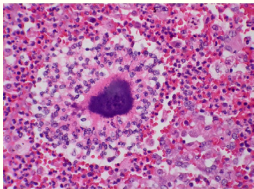
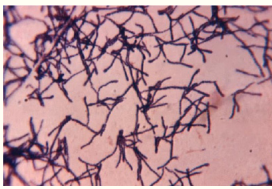
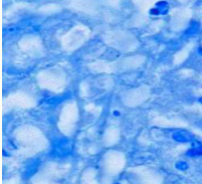
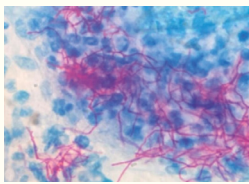
Bacterial Infections

00:01:18

	Staphylococcus aureus	Streptococcus pyogenes (Group - A)	Streptococcus agalactiae (Group - B)
Disease	m/c cause of : • Abscess • Burn wound infection • Native valve IE • Hospital acquired pneumonia (HAP) • Osteomyelitis	• Cellulitis → PSGN • Pharyngitis → ARF	Neonatal sepsis ↓ Neonatal meningitis
Gram stain	 Gram +ve cocci in clusters	 Gram +ve cocci in chains	
Culture	 • β-hemolysis • Large colonies	 • β-hemolysis • Pin-point colonies	
Biochemical tests	Coagulase +ve	Bacitracin sensitive	CAMP test +ve
Rx	• MSSA : Penicillin • MRSA : Vancomycin • VRSA : Daptomycin/Linezolid	Penicillin	

----- Active space -----

	Bacillus anthracis (Zoonotic)	Clostridium perfringens (Anaerobic)
Toxin	Anthrax toxin	Alpha toxin
Disease	- Cutaneous anthrax/Hide Porter's disease : malignant pustule → Black eschar - Pulmonary anthrax/Wool sorter's disease : Hemorrhagic pneumonia	- Gas gangrene : myonecrosis with blebs & bullae - Gastroenteritis : - Diarrhea - Source : Canned food
Gram stain	 Gram +ve bacilli arranged in chains with non-bulging spores Bamboo stick appearance	 Gram +ve bacilli with subterminal bulging spores Club shape appearance
Culture & tests	Blood agar : medusa head/ String of pearls appearance	- RCM culture Nagler's reaction : +ve
Rx	Ciprofloxacin/Doxycycline	Penicillin + Clindamycin

	Actinomyces (Anaerobic)	Nocardia
Disease	Actinomycosis/Lumpy jaw : Jaw : Subcutaneous nodules with discharging granules - Granules : Dark yellow (Similar to sulphur granules)	Actinomycetoma/madura foot : Foot : Subcutaneous nodules with discharging granules - Granules : Whitish-yellow
Gram stain	 Sunray appearance of crushed granule	 Gram +ve, filamentous bacilli
Acid-fast stain	Non-acid fast 	Acid fast 
Rx	Penicillin	Welsh regimen : Amikacin + Cotrimoxazole in cycles

Note :

----- Active space -----

Causes of Actinomycetoma :

- Nocardia (m/c in world).
- Actinomadura (m/c in India).

Mycobacterium leprae :

Non-cultivable.

Leprosy/Hansen's disease :

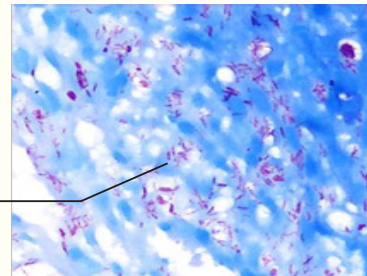
	Paucibacillary (milder)	multibacillary (Severe)
Hypopigmented patches	≤ 5	> 5
Nerve involvement	$\ominus$	≥ 1

Lepra reactions :

	Type 1	Type 2
Features	• Type 4 hypersensitivity reaction • Inflammation of existing lesions	• Type 3 hypersensitivity reaction • Appearance of painful red nodules (Erythema nodosum leprosum)
Rx	• Continue anti-lepromatous therapy • Start steroids	

Investigations :

- Sample : Slit skin smear (Earlobe/elbow).
- microscopy :
 - Acid fast.
 - Non-filamentous.
 - Cigar bundle appearance.
- PCR.
- Animal inoculation : mice/Armadillo.



Rx : Rifampicin + Dapsone + Clofazimine.

DNA Viruses

00:23:35

Human Herpes Viruses (HHV) :

HHV 1, HHV 2 & HHV 3 :

vesicles $\xrightarrow{\text{vesicular fluid}}$ Tzanck smear $\longrightarrow$ multinucleate giant cells (Tzanck cells).

	HHV-1/HSV-1	HHV-2/HSV-2	HHV-3/VZV (Chicken pox virus)
Site of vesicles	upper body	Lower body	Entire body
Other features	Sporadic temporal lobe encephalitis	• Genital herpes $\longrightarrow$ Genital ulcers • Neonatal herpes $\longrightarrow$ mollaret's meningitis	• Primary infection : Chicken pox • Reactivation : Herpes zoster/Shingles

----- Active space -----

Other HHV :

	Virus	Disease
HHV-4	Epstein Barr virus (EBV)	- Infectious mononucleosis (IMN) - Gastric & nasopharyngeal cancer - Burkitt's & Hodgkin's lymphoma
HHV-5	Cytomegalovirus	- IMN - Post-transplant infection (within 3 months)
HHV-6	Human B lymphotropic virus	Encephalitis
HHV-7	RK virus	
HHV-8	KS virus	Kaposi sarcoma : - Reddish, elevated, vascular tumor - m/c in HIV

Investigations :

- PCR.
- Exception : EBV & CMV → ELISA (Ab detection).

Treatment :

- Acyclovir/Ganciclovir.
- Vaccine : Live chickenpox vaccine.

Note :

Post-transplant nephropathy (> 3 months)



BK virus (m/c cause).

Pox Virus :

Clinical features :

- macules.
- umbilicated papules.
- pustules.

Investigation : PCR.

	Small pox	monkey pox	vaccinia	molluscum contagiosum virus
Severity	Severe (Septic shock)	moderate	mild	mild (All papules umbilicated)
vaccine	Live smallpox vaccine			No vaccine

Human Papilloma Virus :

Serotypes :

- Low-risk : 6, 11 → Anogenital warts (Condyloma acuminata).
 - High-risk : 16, 18, 31, 33, 45, 52, 58 }
 - Oropharyngeal cancer
 - urogenital cancer (Penile Ca, cervical Ca)
- Proto-oncogenes :
- E6 : ⊖ p53.
 - E7 : ⊖ RB.

Investigation : PCR.

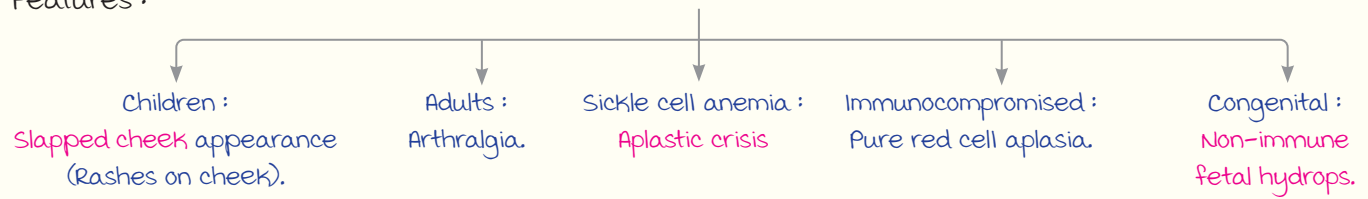
Vaccine : Gardasil - 9.

- Sub-unit : L₁ Ag.
- Age group : 9 - 26 years.

Parvovirus B-19 Serotype :

----- Active space -----

Features :



Investigation : PCR.

RNA Viruses

00:37:35

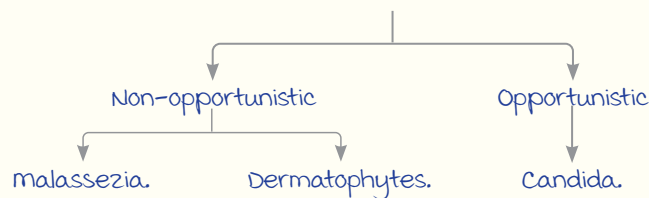
	measles virus	Rubella virus	Coxsackie A-16 & Enterovirus 71
Disease	- Primary infection : - Fever - Koplik spots - Rash - Subacute sclerosing panencephalitis (SSPE) : Reactivation - Seizures - myoclonus - ↓ Scholastic performance	- Fever - Forchheimer spots (Bleeding spots on uvula & hard palate)	Hand-foot-mouth disease (HFMD) : Rashes
Investigations	Ab - ELISA	PCR	
Vaccine	Live : Edmonston-Zagreb strain	Live : RA 27/3	-

Note :

- Coxsackie A-24/Enterovirus 70 → Acute hemorrhagic conjunctivitis.
- Coxsackie B-3 → myocarditis & Bornholm's disease.

Superficial Fungal Infections

00:40:35

malassezia :

- Globosa (m/c) > Furfur.
- Yeast like.

Disease :

Tinea/Pityriasis versicolor :

Hypo/hyperpigmented patches (upper trunk).

Hypopigmented patches

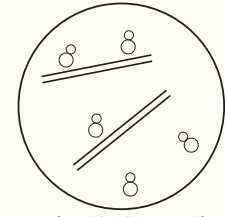


Tinea versicolor

----- Active space -----

Investigations :

- Wood's lamp examination : **Golden yellow fluorescence.**
- KOH wet mount : **Spaghetti & meatball appearance.**
- Urease test : +ve.



Spaghetti & meatball appearance

Treatment :

- Oral : **Itraconazole/Fluconazole.**
- Topical : **Ketoconazole/clotrimazole.**

Dermatophytes :

Septate moulds.

Dermatophytosis/Tinea :

- Tinea capitis (Scalp) : **m. canis** (m/c overall).

Lesion	Involved area	Features	Organism
Kerion	Skin	Boggy lesion (Circular, irregular)	T. verrucosum
Favus	Hair follicle	Cup like crusts (Scutula)	T. schoenleinii
Ectothrix	Outside hair	Grey patch (Alopecia)	m. canis
Endothrix	Inside hair	Grey patch with black dots	T. tonsurans



Kerion



Favus



Ectothrix



Endothrix

- Other sites :
 - **T. rubrum** (m/c) → Red pigment on SDA.
 - Ring worm lesion : Annular, red, scaly, itchy lesion.



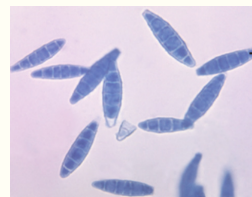
Ringworm lesion

Investigations :

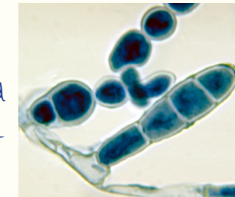
- Wood's lamp examination : **Apple green fluorescence.**
- **Hair perforation test** +ve : **T. mentagrophytes.**
- microscopy :



Trichophyton (m/c)



Microsporum



Epidermophyton

Pencil shaped conidia

Boat shaped conidia

Club shaped conidia

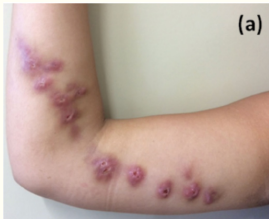
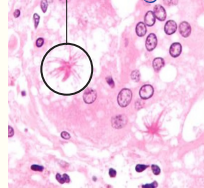
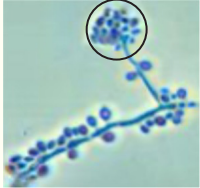
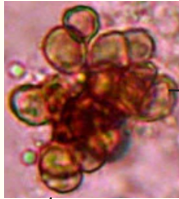
Treatment :

- Oral : **Griseofulvin** (For scalp) or **Itraconazole/Fluconazole** (For other sites).
- Topical : **Ketoconazole/clotrimazole.**

Subcutaneous Fungal Infections

00:53:54

----- Active space -----

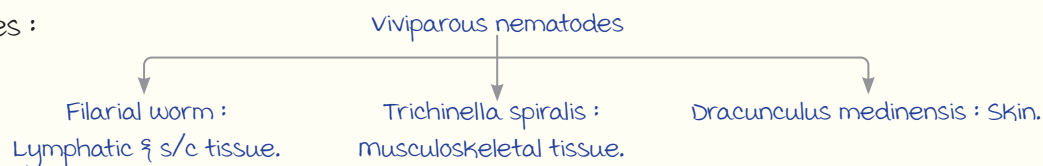
	Features	microscopy
Eumycetoma/ madura foot : madurella mycetomatis	 Subcutaneous (s/c) nodules with discharging granules (Black)	-
Sporotrichosis/Rose Gardner's disease : Sporothrix schenckii	- Subhimalayan region - s/c nodules along lymphatics  (a)	Dimorphic fungus Asteroid body Flower-like conidia  Yeast  mould
Chromoblastomycosis : Cladosporium	Dematiaceous/Pheoid group ↓ - s/c nodules & verrucous (Wart-like) lesions - Brown/black pigment on SDA	 Copper penny appearance medlar/sclerotic bodies
Pheohyphomycosis : Bipolaris		medlar bodies ⊖

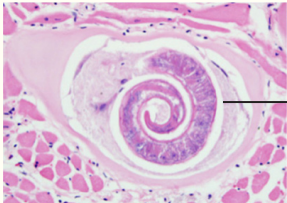
Treatment : Itraconazole (Oral)/Lipo-Amphotericin B (IV).

Parasitic Infections

01:01:20

Causes :



	Trichinella spiralis (Spiral worm)	Dracunculus medinensis (Guinea worm)
Host	man/Pig	man/Cyclops
c/f	- myalgia - myocarditis	Skin blisters
Diagnosis	- muscle biopsy :  Spiral larva - Bachman test +ve	Adult worm from ruptured blisters 
Rx	Albendazole	Worm removal

MISCELLANEOUS INFECTIONS

Bacterial Toxin Related Diseases 🚑

00:00:52

mechanisms of action :

↑ cAMP

⊖

Rho-GTPase

⊖

Neurotransmitters
(NT)

⊖

Protein
synthesis

⊖

Elongation
factor 2 (EF-2)

⊖

Ribosomes

Bacterial toxins :

Toxin	MOA	Disease	Treatment
Anthrax toxin (Bacillus anthracis)	↑ cAMP	- Cutaneous anthrax/Hide porter's disease - Pulmonary anthrax/Wool sorter's disease	Ciprofloxacin
Cholera toxin (Vibrio cholerae)	Binds to G_{M1} ganglioside receptor (In intestine) ↓ Disruption of zona occludens ↓ ↑ cAMP	Cholera : Rice watery diarrhea	Doxycycline
Exotoxin A & B (Pathogenic Clostridium difficile)	⊖ Rho GTPase	Pseudomembranous colitis	Fidaxomicin
Botulinum toxin (Clostridium botulinum)	⊖ Excitatory NT : Acetylcholine	Botulism : Flaccid paralysis	Antitoxin + metronidazole
Tetanospasmin (Clostridium tetani)	⊖ Inhibitory NT : GABA & glycine	Tetanus : Spastic paralysis	
Diphtheria toxin (Corynebacterium diphtheriae)	⊖ EF-2	Diphtheria : - Tonsillar pseudomembrane - Bull neck lymphadenopathy	Antitoxin + Penicillin
Exotoxin A (Pseudomonas aeruginosa)		- Pneumonia (In cystic fibrosis) - Contact lens keratitis	Ceftazidime + meropenem
Verocytotoxin/Shiga like toxin (EHEC O157 : H7)	⊖ Ribosomes	- Dysentery - Hemolytic uremic syndrome	Ciprofloxacin
Shiga toxin (Shigella)			

Note :

C. difficile (Intestinal commensal) $\xrightarrow[\text{Ceftriaxone/Ampicillin/Clindamycin}]{\text{Prolonged use of antibiotics :}}$ *C. difficile* (Pathogen).

----- Active space -----

Bacterial Zoonotic Diseases

00:08:38

Source	Organism	Zoonoses	manifestation	Treatment
Cattle	<i>Bacillus anthracis</i>	Anthrax	Black eschar > Hemorrhagic pneumonia	Ciprofloxacin
	<i>Brucella</i>	Brucellosis	- Hepatosplenomegaly - myalgia - Arthralgia	Doxycycline + Rifampicin
Rat	<i>Yersinia pestis</i>	Plague	Buboes > Pneumonia	Streptomycin/ Doxycycline
	<i>Leptospira</i>	Leptospirosis	- Hepatosplenomegaly - Calf tenderness ⊕ - Conjunctival suffusion	Doxycycline/ Penicillin
Horse	<i>Burkholderia mallei</i>	Glander's disease	Pneumonia	Ceftazidime + meropenem

Bacterial Vector Transmitted Diseases

00:11:39

DOC : Doxycycline.

- Exception : *Yersinia pestis* (Streptomycin > Doxycycline).

Organism	Species	vector	Disease	manifestation
<i>Yersinia</i>	<i>Yersinia pestis</i>	Rat flea (<i>Xenopsylla cheopis</i>)	Plague	Bubo > Pneumonia
<i>Borrelia</i>	<i>Borrelia burgdorferi</i>	Hard tick (<i>Ixodid tick</i>)	Lyme's disease	Target rash/ erythema migrans
	<i>Borrelia recurrentis</i>	Louse	Epidemic relapsing fever	Recurrent fever
	<i>Borrelia duttoni</i>	Soft tick (Argasid tick)	Endemic relapsing fever	
Rickettsia family	<i>Rickettsia prowazekii</i>	Louse	Epidemic typhus	- Fever - Rash - myalgia
	<i>Rickettsia typhi</i> /mooseri	Flea	Endemic typhus	
	<i>Orientia tsutsugamushi</i>	mite	Scrub typhus (Black eschar)	
	<i>Rickettsia rickettsii</i>	Tick	Rocky mountain spotted fever	
	<i>Rickettsia conorii</i>		Mediterranean spotted fever	
	<i>Rickettsia akari</i>	mite	Rickettsial pox spotted fever	
	<i>Coxiella burnetii</i>	Soft tick (m/c route : Inhalation)	Q fever	
	<i>Bartonella quintana</i>	Louse	Trench fever	

----- Active space -----



Rat flea



Louse



Soft tick



Hard tick



mite

Congenital Infections

00:22:01

mnemonic : **TORCH** infections.

Organism		manifestations
Toxoplasma		Triad : • Eye : Chorioretinitis • Brain : - Hydrocephalus - Diffuse calcifications
Others	Treponema pallidum	Syphilitic triad : • Eye : Keratitis • Ear : Sensorineural hearing loss • Teeth : Hutchinson's teeth
	Zika virus	• microcephaly • Subcortical calcifications
	Varicella zoster virus	• Cicatrization • Limb hypoplasia
	Parvovirus	• Non immune fetal hydrops
Rubella (m/c in 1 st trimester)		Triad : • Eye : Cataract • Ear : Sensorineural hearing loss • Heart : Patent ductus arteriosus
Cytomegalovirus (Overall m/c)		Periventricular calcifications
Herpes simplex virus		Neonatal herpes

Prion Disease

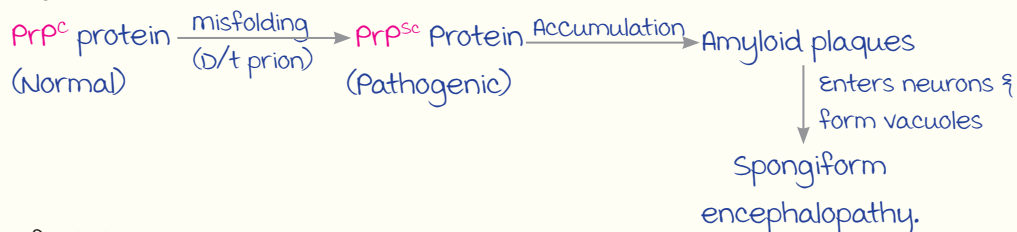
00:24:33

----- Active space -----

Creutzfeldt - Jakob Disease (CJD) :

Etiological agent : Prion (Infectious protein).

Pathogenesis :



manifestations :

- Progressive mental deterioration.
- myoclonus.
- Dementia.

Investigations :

- CSF ELISA :
 - To detect PrP^{sc} protein.
 - Confirmatory test.
- MRI : Cortical ribboning.
- EEG : Triphasic high voltage discharges.

Bioterrorism Agents & Biosafety Levels

00:28:47

Categories	Characteristics	Bioterrorism agents/diseases	Biosafety levels
Category A	• Spread easily • High mortality	Bacteria : • Anthrax • Botulism • Plague	BSL-3
		Viruses : • Smallpox • Ebola • Marburg	BSL-4
Category B	moderate spread & mortality	Bacteria causing gastroenteritis	BSL-2
Category C	• Spread easily • High mortality • Emerging pathogens • Bioengineered for mass dissemination	• Virus : - Nipah - H₁N₁ - Corona • Bacteria : - Mycobacterium tuberculosis	BSL-3

BSL-1 : Non pathogenic organisms.

